



Programmable and Energy-Efficient Hardware Acceleration for AI, Cryptography, and Quantum Computing

PhD. Pham Hoai Luan

Introduction



Myself:

- Name: Phạm Hoài Luân.
- Age: 30.
- Email: pham.luan@is.naist.jp or luanph@uit.edu.vn



Education History:

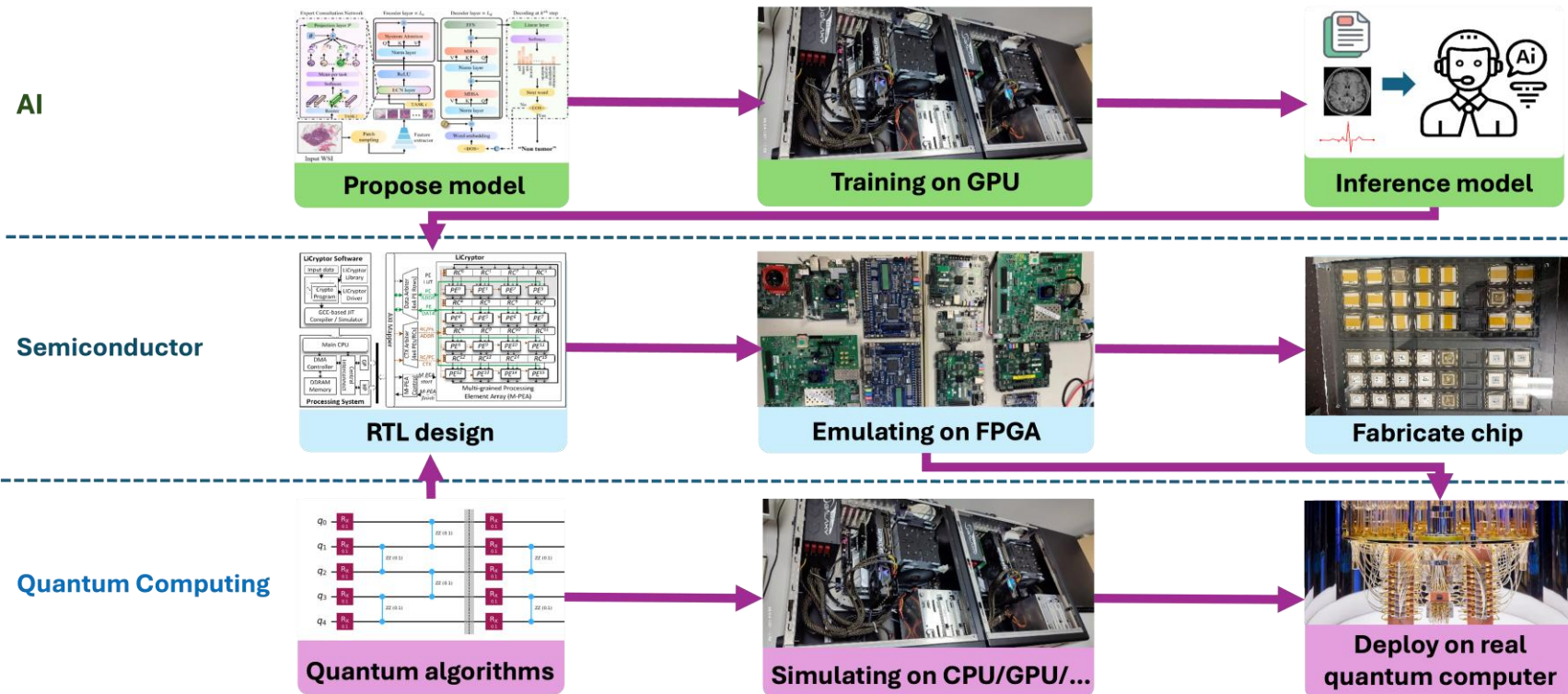
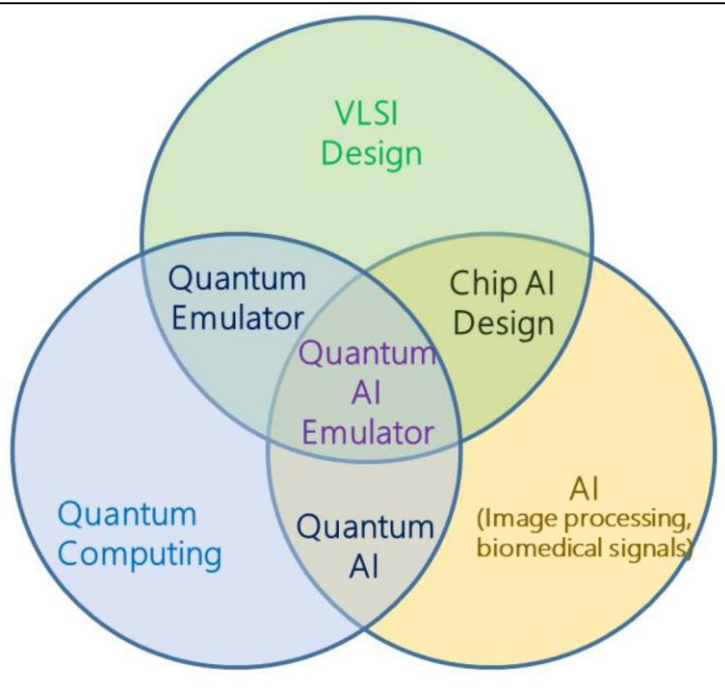
- 2013-2018: Bachelor at UIT-VNU, Vietnam
- 2018-2020: Master's degree at Nara Institute of Science and Technology (NAIST), Japan.
- 2020-2022: PhD at NAIST, Japan.



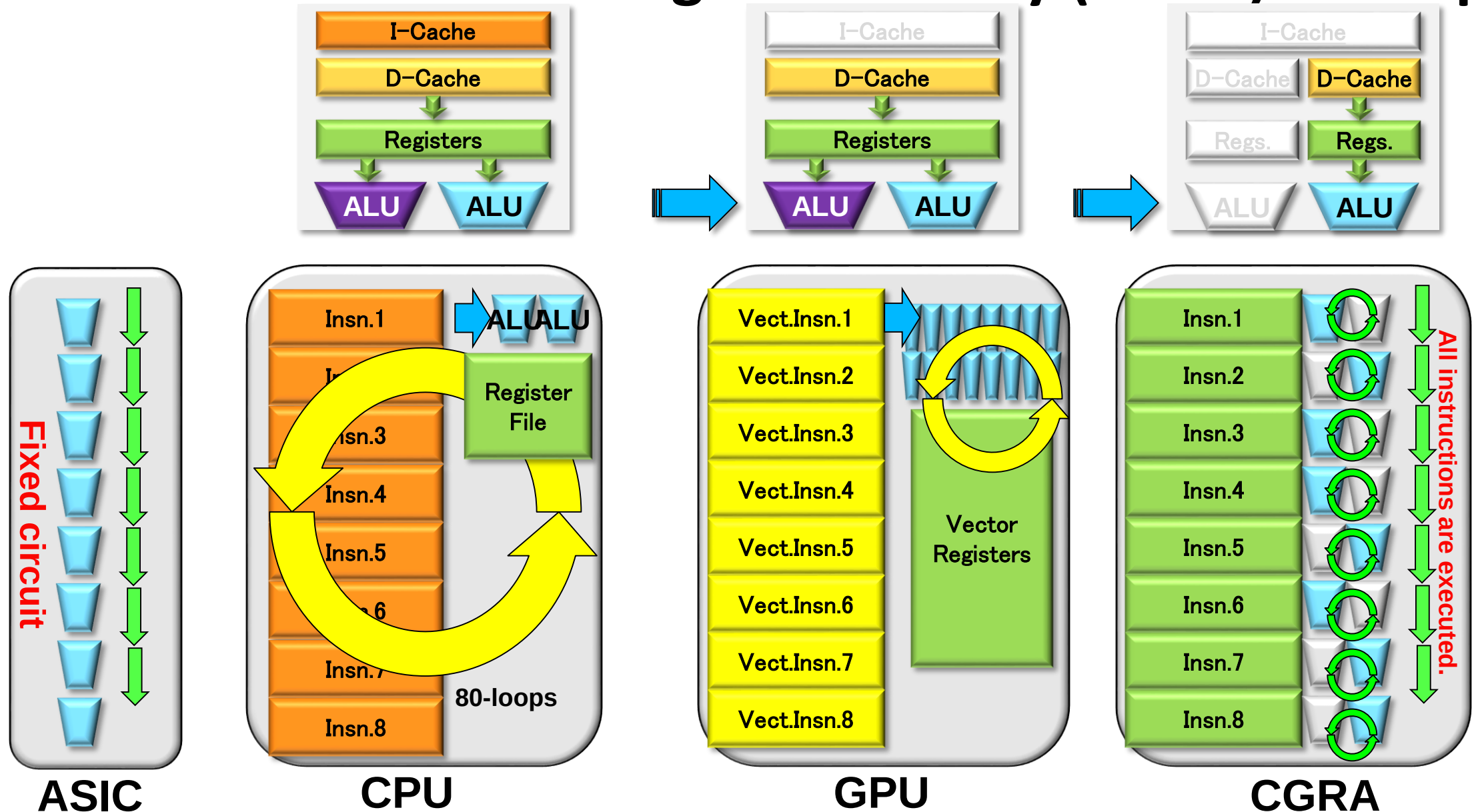
Work History:

- 2022/10-now: Assistant Professor at NAIST, Japan.
- 2022/11-now: Lecturer at UIT-VNU, Vietnam.
- 2025/01-now: Visiting Lecturer at VJU-VNU, Vietnam.
- 2025/04-now: SoC Engineer at LENZO company, Japan
- 2025/08-now: Leader/Founder at AISeQ Lab, Vietnam.

AISeQ Lab Vision



Coarse-Grained Reconfigurable Array (CGRA) Concept

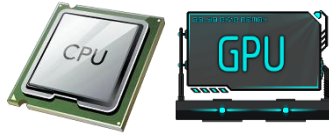


Coarse-Grained Reconfigurable Array (CGRA) Concept



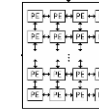
ASIC

- ✓ High Speed
- ✓ Small Area
- ✓ Low Power
- ❖ Low Flexibility



CPU/GPU

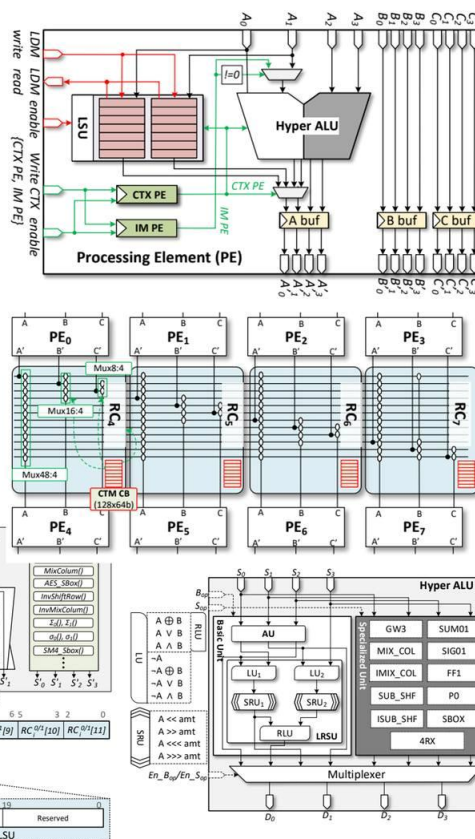
- ✓ Moderate/High Speed
- ❖ Large area
- ❖ High Power
- ✓ High Flexibility



CGRA

- ✓ High Speed
- ✓ Small Area
- ✓ Low Power
- ✓ High Flexibility

I. CGRA-based Accelerator for Cryptography



- ☐ **MRCA 2.0: An Area-Optimized Multigrained Reconfigurable Cryptographic Accelerator for Securing Blockchain-Based Internet of Things Systems** 

Hoai Luan Pham; Vu Trung Duong Le; Tuan Hai Vu; Van Duy Tran; Van Tinh Nguyen; Thi Diem Tran; Yasuhiko Nakashima

IEEE Micro

Year: 2025 | Volume: 45, Issue: 2 | Magazine Article | Publisher: IEEE

☐ Abstract [HTML](#)  

☐ **CTFE: A High-Efficient Heterogeneous Cryptographic CGRA for Diverse Security Applications** 

Vu Trung Duong Le; Hoai Luan Pham; Thi Hong Tran; Van Duy Tran; Tuan Hai Vu; Yasuhiko Nakashima

IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems

Year: 2025 | Volume: 44, Issue: 5 | Journal Article | Publisher: IEEE

Cited by: [Papers \(2\)](#)

☐ Abstract [HTML](#)  

☐ **LiCryptor: High-Speed and Compact Multi-Grained Reconfigurable Accelerator for Lightweight Cryptography** 

Hoai Luan Pham; Vu Trung Duong Le; Van Duy Tran; Tuan Hai Vu; Yasuhiko Nakashima

IEEE Transactions on Circuits and Systems I: Regular Papers

Year: 2024 | Volume: 71, Issue: 10 | Journal Article | Publisher: IEEE

Cited by: [Papers \(5\)](#)

☐ Abstract [HTML](#)  

☐ **Flexible and Scalable BLAKE/BLAKE2 Coprocessor for Blockchain-Based IoT Applications** 

Hoai Luan Pham; Thi Hong Tran; Vu Trung Duong Le; Nakashima Yasuhiko

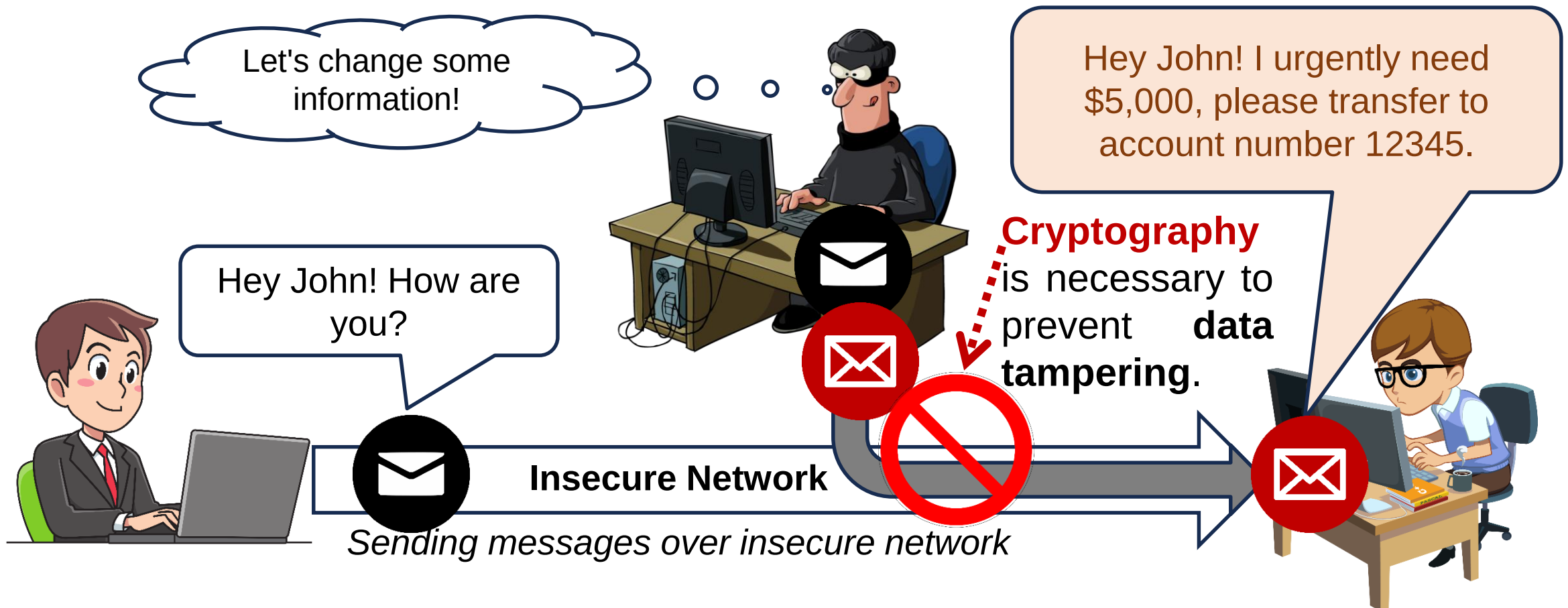
IEEE Design & Test

Year: 2023 | Volume: 40, Issue: 5 | Magazine Article | Publisher: IEEE

Cited by: [Papers \(9\)](#)

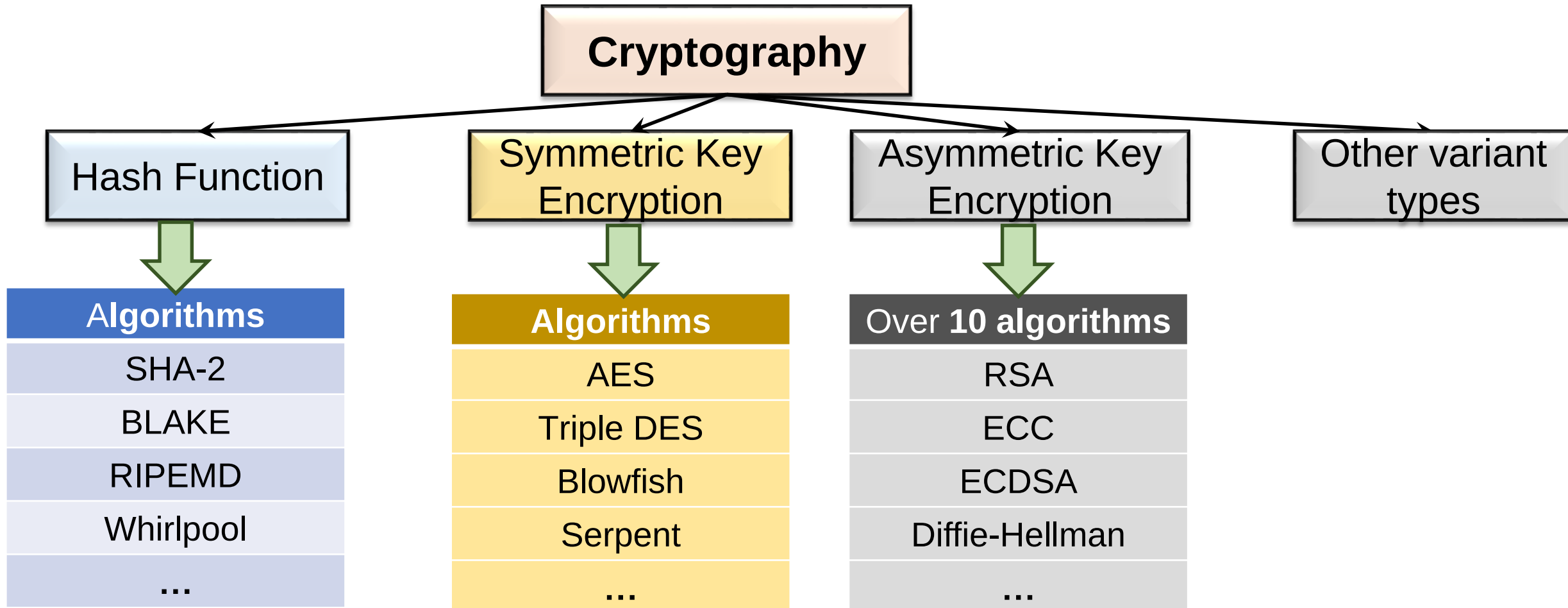
Background

- Nowadays, cryptography is used to prevent hackers from **changing or tampering** with important information.



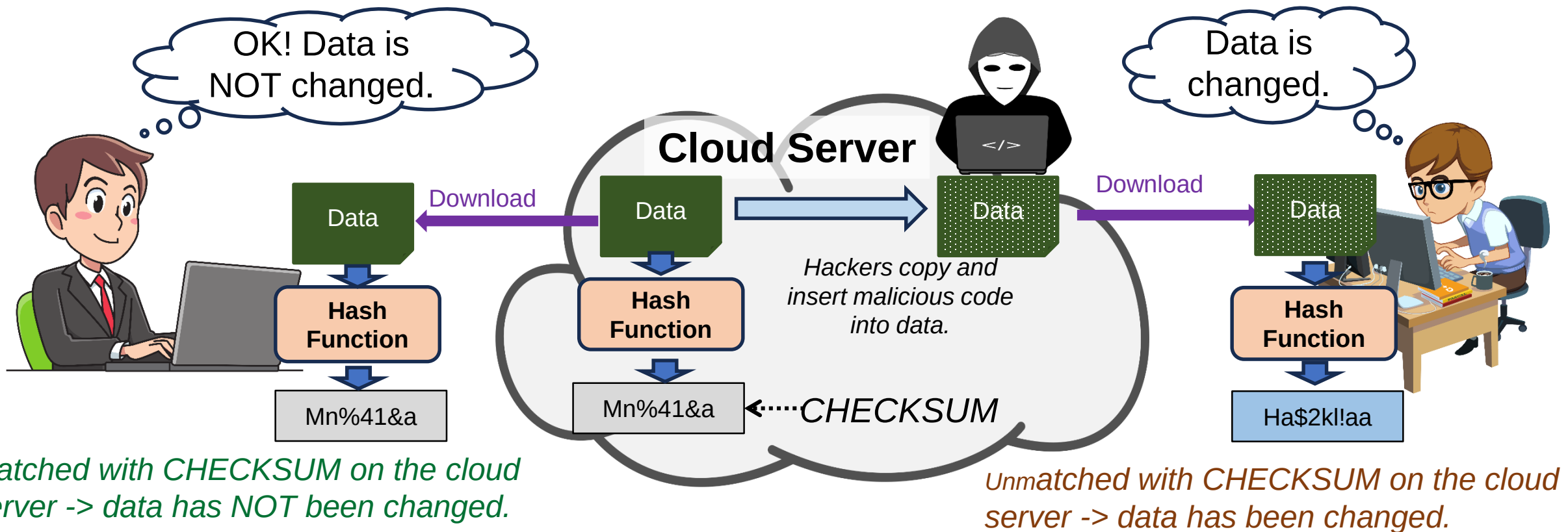
Background

- ❑ There are three main types of cryptography: **hash function**, **symmetric key encryption**, and **asymmetric key encryption**.



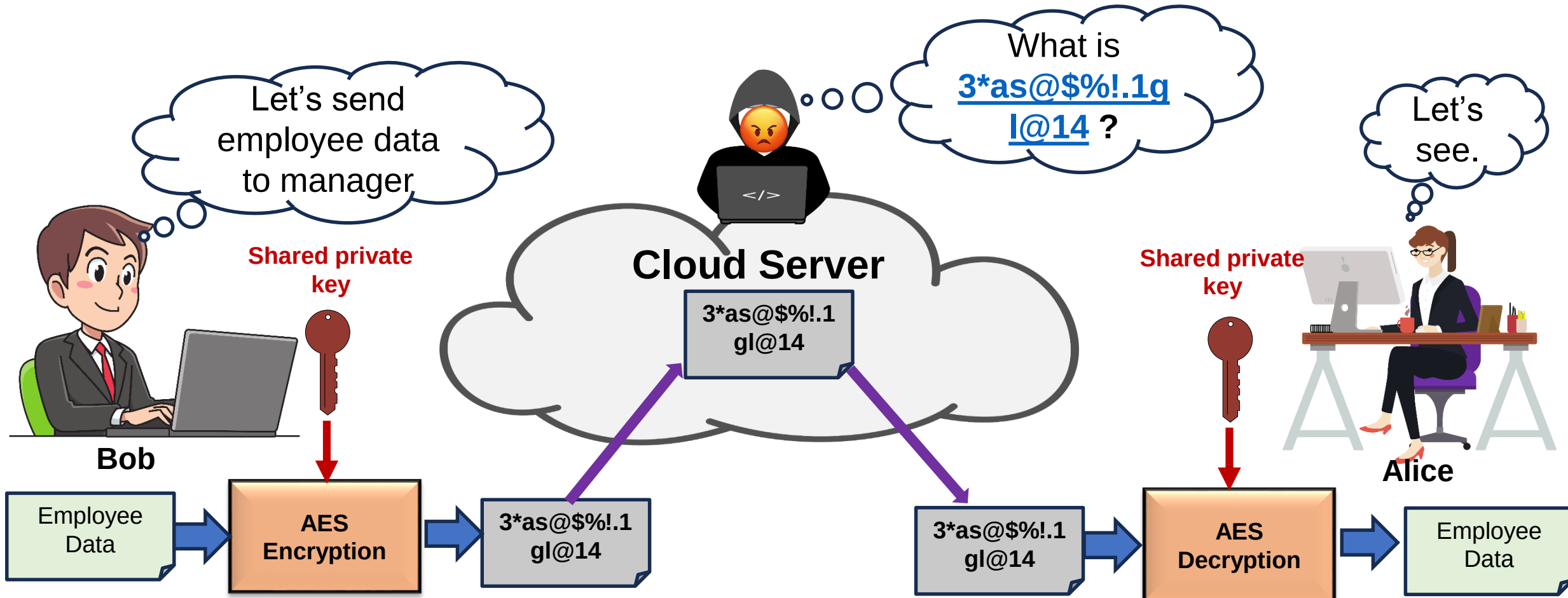
Background

- ❑ Hash function can be used to **check data integrity** by generating checksums, ensuring that any alterations to the data are detectable.



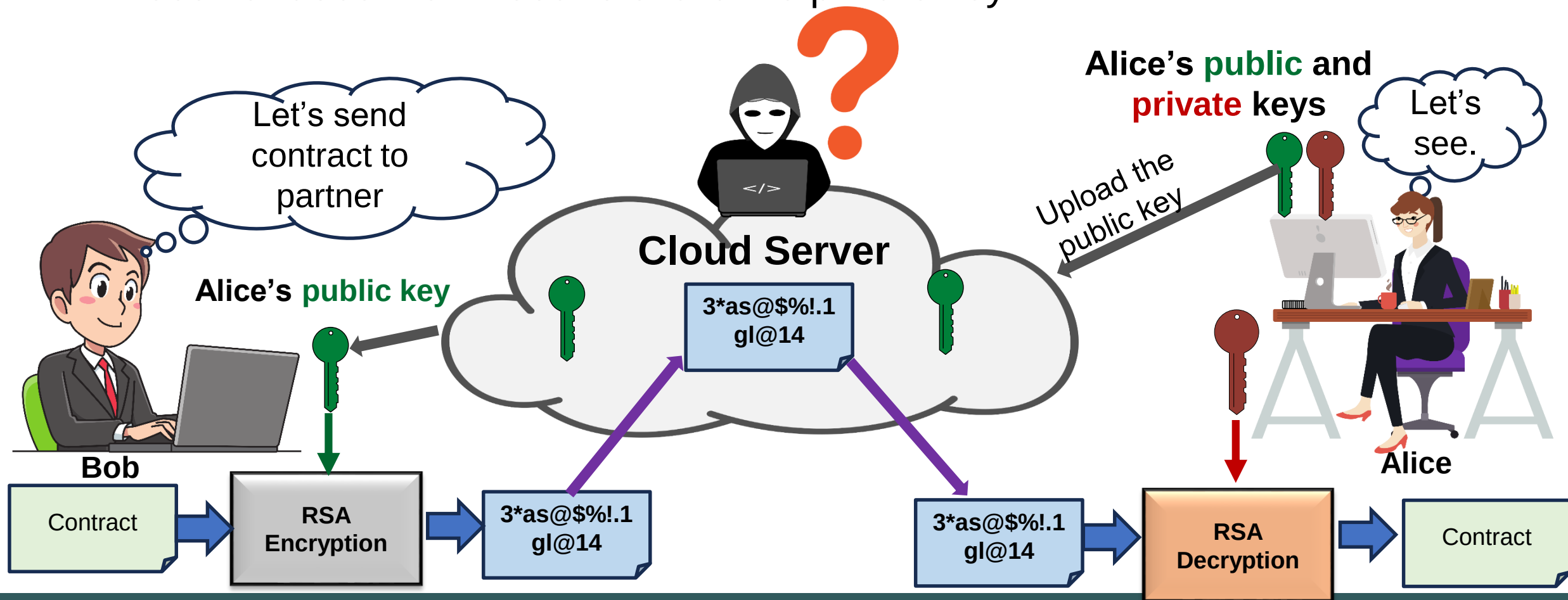
Background

- ❑ **Symmetric key encryption** is used for encrypting **sensitive data stored** in databases, file systems, and cloud server to protect against unauthorized access (hacker).

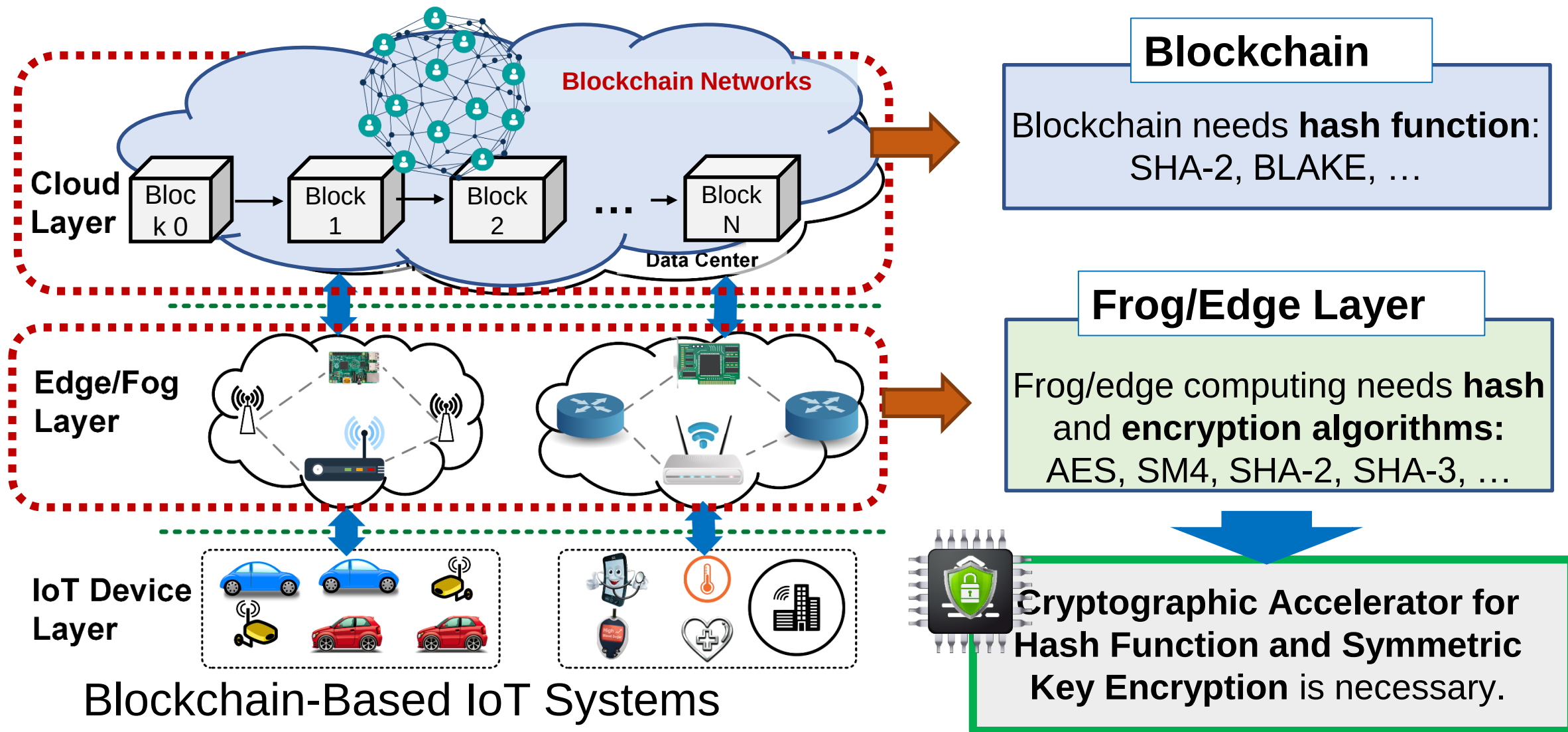


Background

- ❑ **Asymmetric key encryption** is also used to encrypt **sensitive data** during transmission to protect against unauthorized access (hackers), where the receiver does NOT need to share the private key.

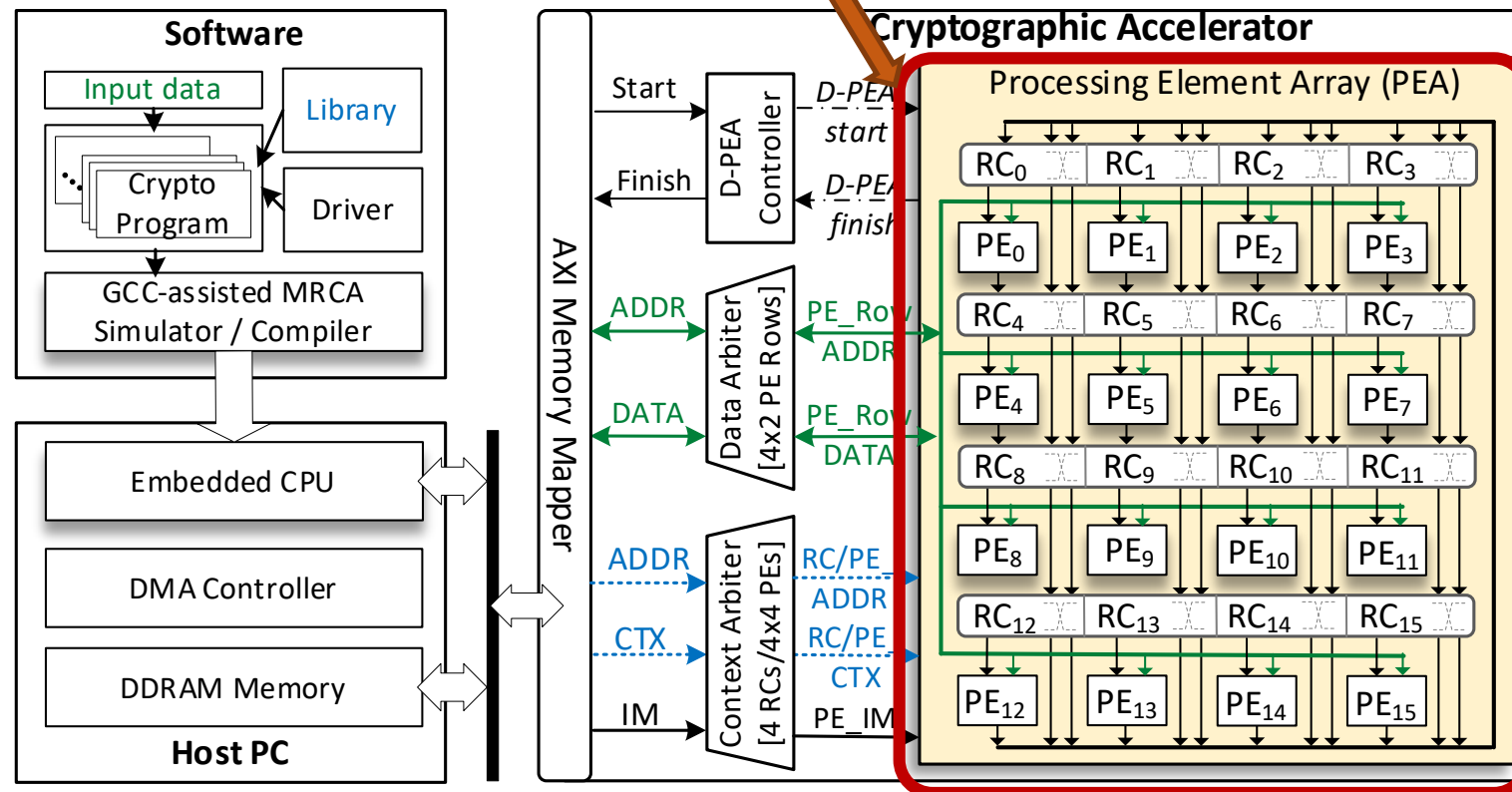


Background



Proposed Cryptographic Accelerator

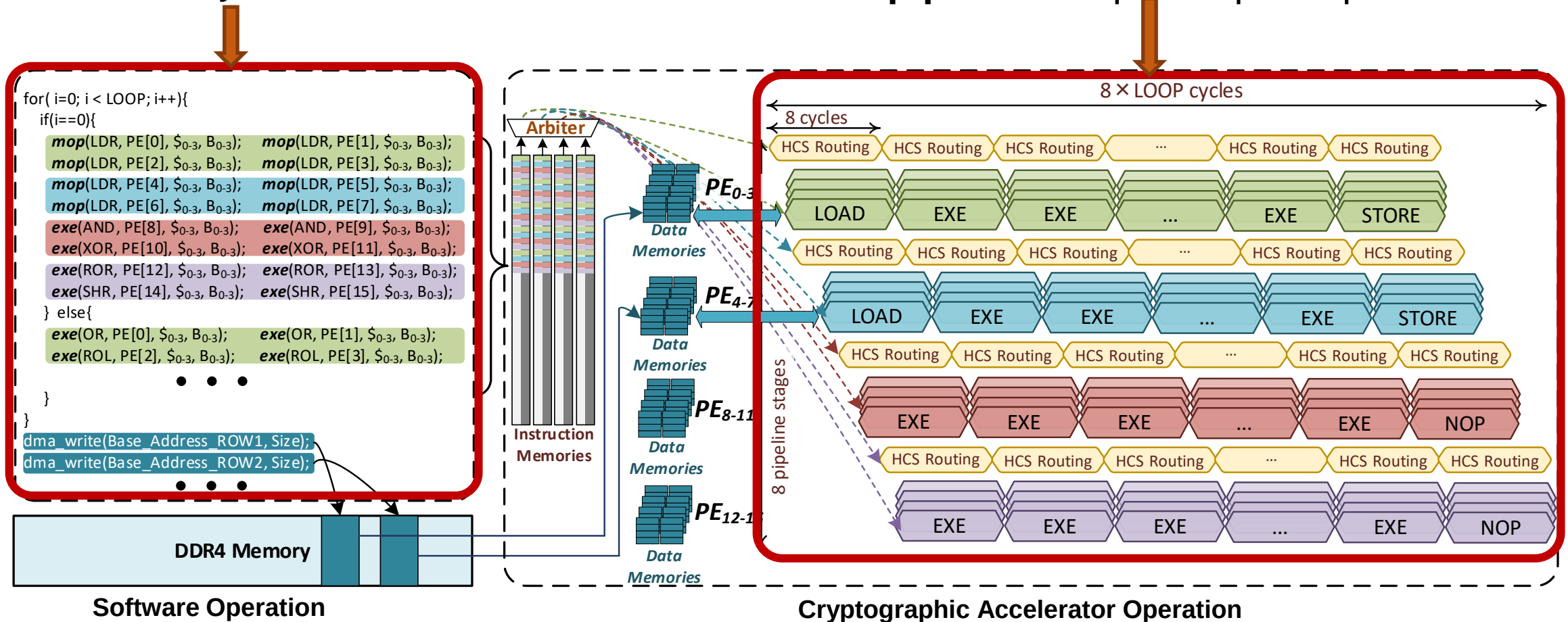
- ❑ The **PEA** enhances speed and flexibility through dynamic reconfiguration for various cryptographic algorithms



Why High Flexibility and Performance?

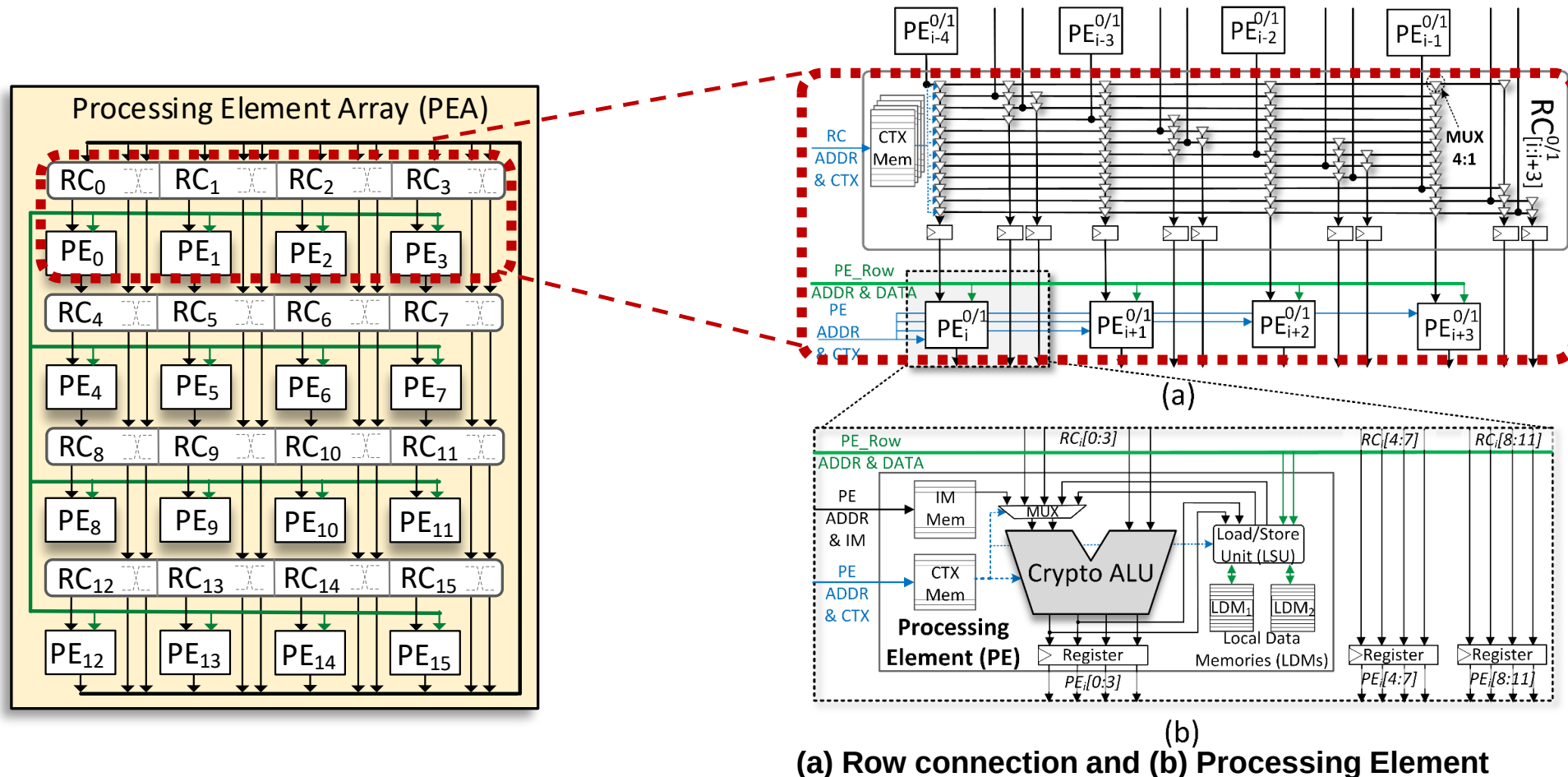
- ❑ The PEA is **dynamically configurable** to change different algorithms but at the assembly level

- ❑ The PEA in the Cryptographic Accelerator operates in a **deep pipeline** to speed up computation.



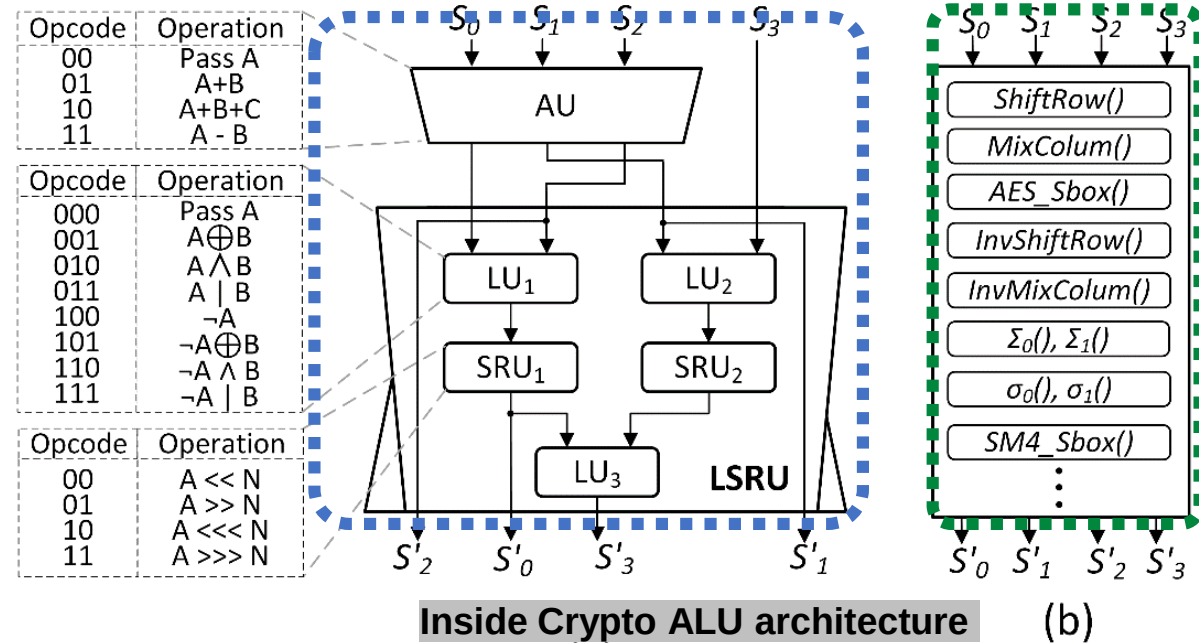
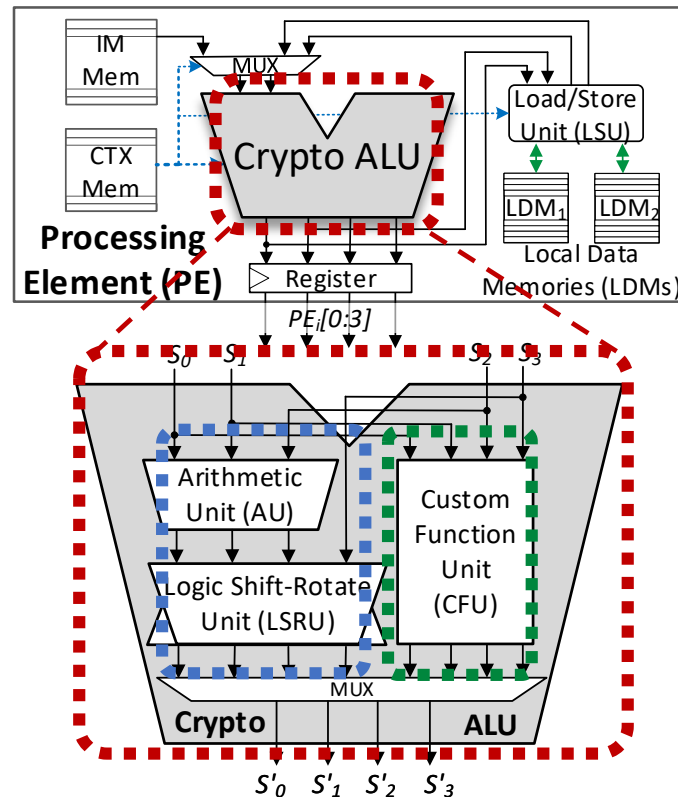
Why High Flexibility and Performance?

- In the PEA architecture, the row connection (RC) is **reconfigurable for flexible data routing**, and the processing element (PE) can **perform any cryptographic operation**, thereby achieving high flexibility.



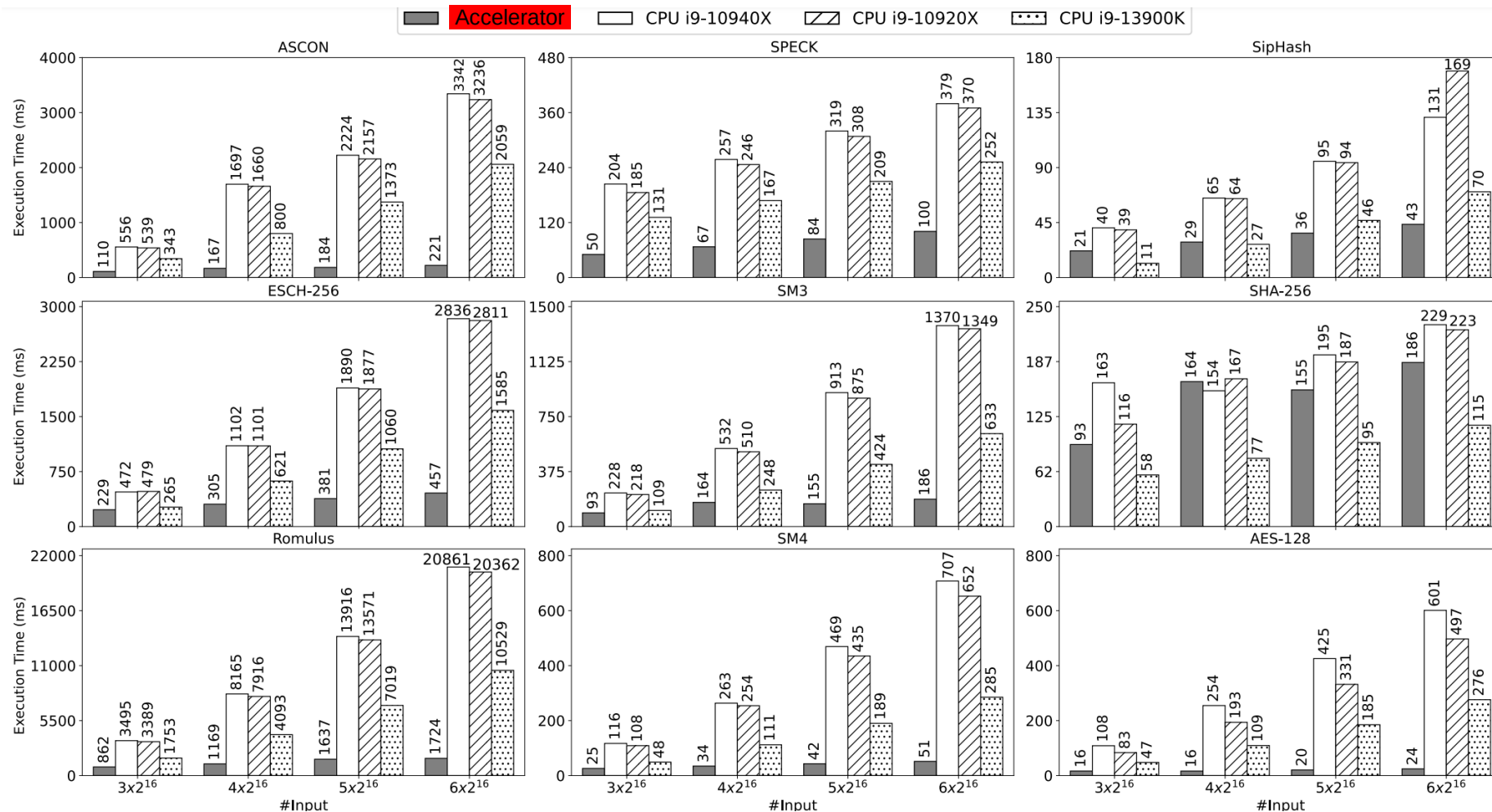
Why Low Power?

- ❑ The Crypto ALU inside each PE can compute **multiple operators in a single cycle** to significantly enhance performance.
- ❑ The Crypto ALU contains **only simple and low-cost operators, resulting in very low power consumption.**



Result

✓ 4 times speed up

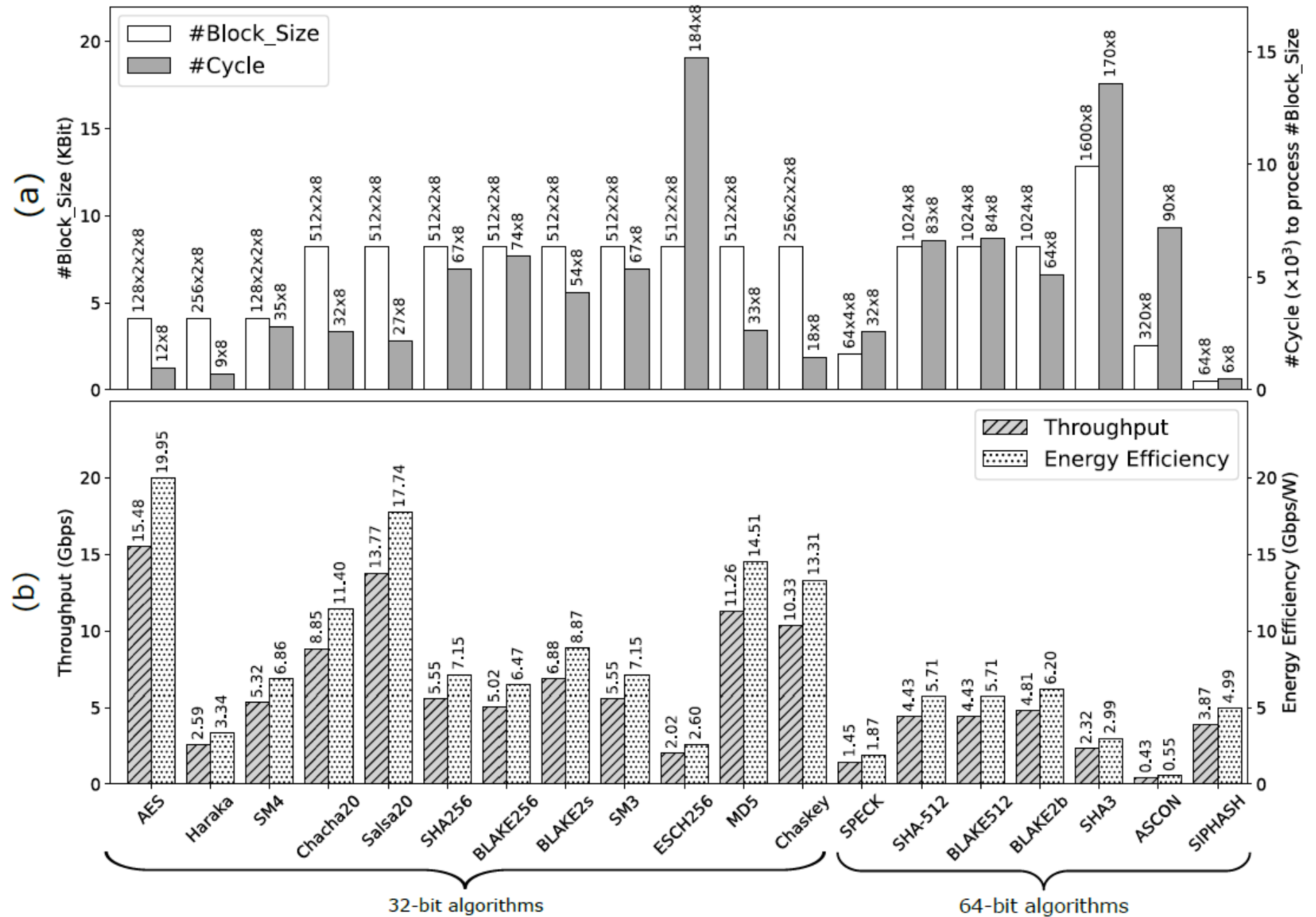


Execution time between Cryptographic Accelerator and strongest CPUs

Result

✓ Support up to 19 algorithms

✓ Throughput: 0.43-15.48 Gbps



Result

✓ 1.29-1.59 times better in Energy Efficiency

Reference		ICCAD'14 [29]	TCAD'18 [30]	TCAS-II'20 [31]	TVLSI'20 [28]	IEICE'23 [32]	KSE'23 [34]	Our LiCryptor
Architectural Type		CGRA	CGRA	CGRA	CGRA	CGRA	CGRA	CGRA
Key Granularity Support		8/32	8/32	8/32	8/32	32	32	8/32/64
Technology (nm)		45	65	65	55	65	45	45
Frequency (MHz)		1000	400	500	130	170	254	417
Area (mm ²)		6.32	7.75	9.91	12.25	19	5.67	2.29
TP (Gbps)	ASCON	✗	—	—	✗	✗	✗	1.570
	SPECK	✗	✗	✗	✗	✗	✗	1.570
	AES-128	128	51.2 / 73.96 [†]	64 / 92.16 [†]	0.44 / 0.54 [†]	✗	✗	8.896
	SHA256	1.60	0.46 / 0.66 [†]	0.80 / 1.16 [†]	0.38 / 0.46 [†]	1.32 / 1.9 [†]	4.06	3.140
	SM3	—	0.64 / 0.92 [†]	0.66 / 0.95 [†]	—	—	—	3.140
AE (Gbps/mm ²)	ASCON	✗	—	—	✗	✗	✗	0.69
	SPECK	✗	✗	✗	✗	✗	✗	0.69
	AES-128	3.28	6.61 / 12.96 [†]	6.46 / 12.66 [†]	0.04 / 0.05 [†]	✗	✗	3.88
	SHA256	0.25	0.06 / 0.12 [†]	0.08 / 0.16 [†]	0.03 / 0.05 [†]	0.07 / 0.14 [†]	0.72	1.37
	SM3	—	0.08 / 0.16 [†]	0.07 / 0.13 [†]	—	—	—	1.37

[28] Y. Du, W. Li, Z. Dai, and L. Nan, "Pvharrray: An energy-efficient reconfigurable cryptographic logic array with intelligent mapping," IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 28, no. 5, pp. 1302–1315, 2020.

[29] G. Sayilar and D. Chiou, "Cryptoraptor: High throughput reconfigurable cryptographic processor," in 2014 IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2014, pp. 155–161.

[30] L. Liu, B. Wang, C. Deng, M. Zhu, S. Yin, and S. Wei, "Anole: A highly efficient dynamically reconfigurable crypto-processor for symmetric-key algorithms," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 37, no. 12, pp. 3081–3094, 2018.

[31] C. Deng, B. Wang, L. Liu, M. Zhu, Y. Wu, H. Li, S. Yin, and S. Wei, "A 60 gb/s-level coarse-grained reconfigurable cryptographic processor with less than 1-w power," IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 67, no. 2, pp. 375–379, 2020.

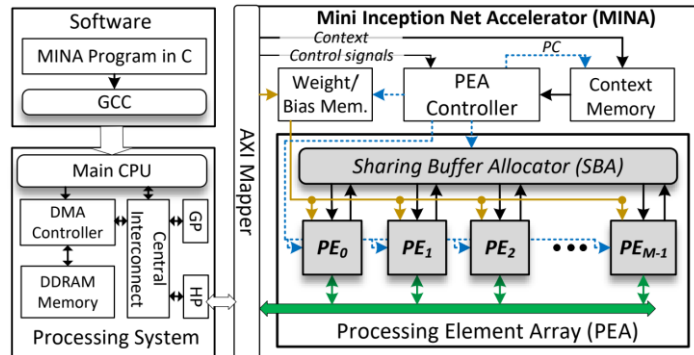
[32] V.-T.-D. LE, H.-L. PHAM, T.-H. TRAN, and Y. NAKASHIMA, "Flexible and energy-efficient crypto-processor for arbitrary input length processing in blockchain-based iot applications," IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences, vol. advpub, p. 2023CIP0013, 2023.

[34] V. T. Duong Le, H. L. Pham, T. S. Duong, T. H. Tran, Q. D. Nam Nguyen, and Y. Nakashima, "Rhcp: A reconfigurable high-efficient cryptographic processor for decentralized iot platforms," in 2023 15th International Conference on Knowledge and Systems Engineering (KSE), 2023, pp. 1–6.

II. CGRA-based Accelerator for AI

Publication

Deep Neural Accelerator for Biosignal Processing



MINA: A Hardware-Efficient and Flexible Mini-InceptionNet Accelerator for ECG Classification in Wearable Devices

Hoai Luan Pham; Thi Diem Tran; Vu Trung Duong Le; Yasuhiko Nakashima
IEEE Transactions on Circuits and Systems I: Regular Papers
Year: 2025 | Volume: 72, Issue: 6 | Journal Article | Publisher: IEEE
Cited by: Papers (1)

A Fast and Memory-Efficient CNN Accelerator for ECG Classification in Remote Healthcare Systems

Hoai Luan Pham; Thi Diem Tran; Vu Trung Duong Le; Tuan Hai Vu; Yasuhiko Nakashima
2025 22nd International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)
Year: 2025 | Conference Paper | Publisher: IEEE

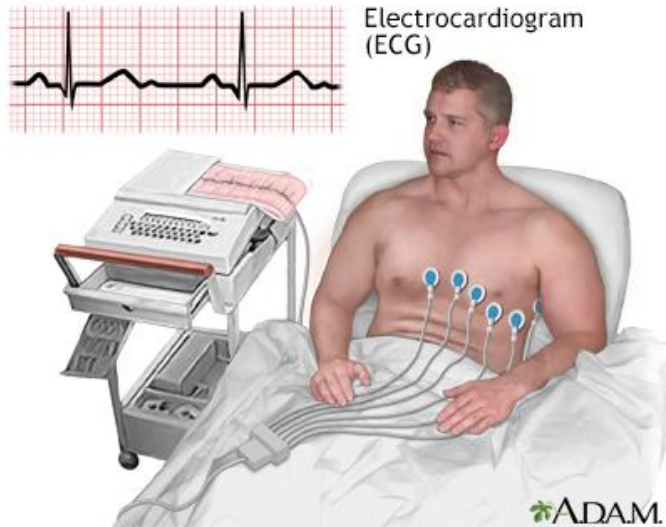
A High-Performance FPGA Based on Convolutional Neural Network for QRS Complex Detection

Binh Tan Ngo; Nghi Hoang Huu Thai; Hoai Luan Pham; Thi Diem Tran
2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)
Year: 2024 | Conference Paper | Publisher: IEEE

Reference	[18]	[19]	[20]	[21]	[22]	[23]	[24]	This Work
Model	2-D CNN	2-D CNN	2-D CNN	2-D CNN	2-D CNN	2-D CNN	2-D CNN	1-D CNN
Dataset	CinC	-	CinC	CinC	CinC	MIT-BIH	MIT-BIH	MIT-BIH
Application	ECG, others*	ECG, others*	ECG rhythm	ECG rhythm	ECG rhythm	ECG beat	ECG beat	ECG beat
FPGA	Z-7045	ZC702	Z-7045	ZC702	ZC702	PYNQ-Z1	ZCU104	ZCU102 / ZC706
Clock (MHz)	150	120	120	120	100	100	100	250 / 200
CNN Size (GOP)	-	-	-	-	-	-	-	0.32768×10^{-3}
Parameter	50.12M	-	60,840	-	-	3.23M	30,573	6,457
Precision	16-bit Fixed	16-bit Fixed	16-bit Fixed	16-bit Fixed	8-bit Fixed	8-bit Fixed	16-bit Fixed	16-bit Fixed
Area	LUT	182,616	38,136	99,546	42,964	25,057	17,579	51,548
	BRAM	486	242	320	128	133	85	133
	DSP	780	205	864	220	151	85	133
	eLUT [†]	782,040	285,264	592,346	204,916	171,609	1,118,595	200,508
Throughput (GOP/s)	117.0	41.0	61.91	26.5	15.10	13.3	8.0	9.85 / 7.88
EE (GOP/s/MeLUT)	149.61	143.73	104.52	129.32	87.99	75.99	66.33	249.92 / 191.99

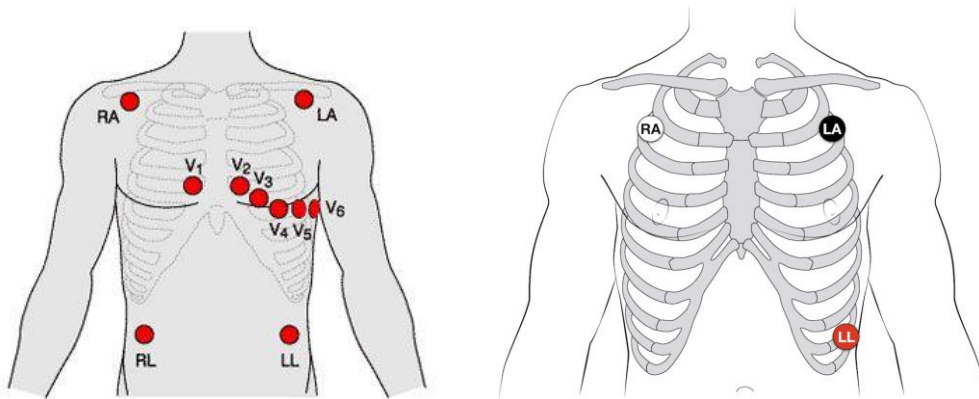
* The application includes other tasks beyond ECG; [†] The eLUT is normalized by #LUT + #DSP $\times$ 280 + #BRAM $\times$ 784 [35].

Background



Source: <https://www.mountsinai.org/health-library/tests/electrocardiogram>

- ECGs monitor heart activity and are vital for early detection of cardiac ischemia, a leading global cause of death, with **17.8 million deaths** in 2017 (WHO).
- ECGs are used in both **12-lead** hospital systems and increasingly in **3-lead** wearable devices.



12-lead ECG

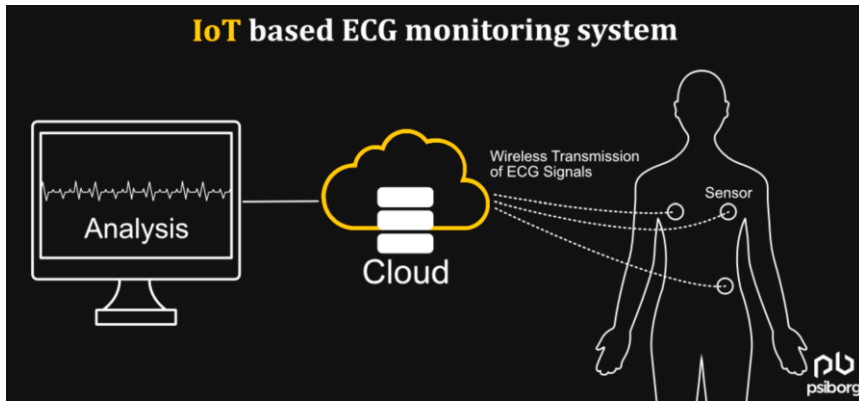
3-lead ECG

Source: <https://www.firstaidforfree.com/recording-a-12-lead-ecgkg/>

Source: <https://litfl.com/ecg-lead-positioning/>

- **Hospital ECGs** provide high-resolution data but are **uncomfortable** and not ideal for long-term use, while **wearable ECGs** enable **continuous monitoring** with smart features like real-time arrhythmia detection.

Background



Source: <https://psiborg.in/iot-based-ecg-monitoring-system-for-remote-healthcare/>

- IoT-based ECG monitoring is emerging as a solution to **reduce hospital burden** and enable continuous remote care.

- Accurate ECG classification in remote settings requires **AI/ML algorithms** that are both power- and compute-efficient.

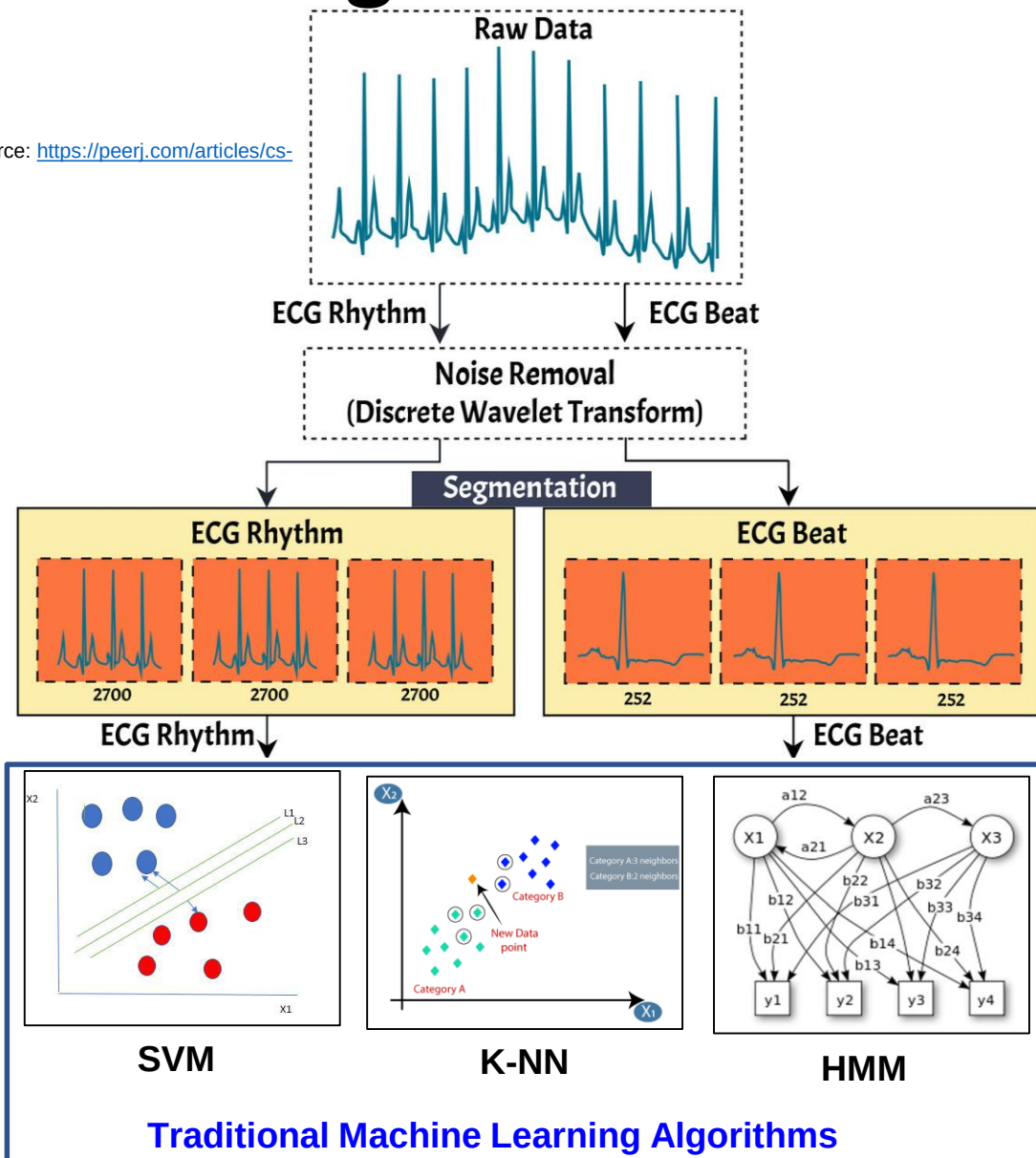
- **Fast and energy-efficient ECG hardware** is essential for next-generation wearable healthcare devices.



Source: <https://www.businesswire.com/news/home/20211021005727/en/Proliferation-of-Remote-Health-Monitoring-Fueled-by-Post-COVID-Era-Finds-Strategy-Analytics>

Background

Source: <https://peerj.com/articles/cs-825/>



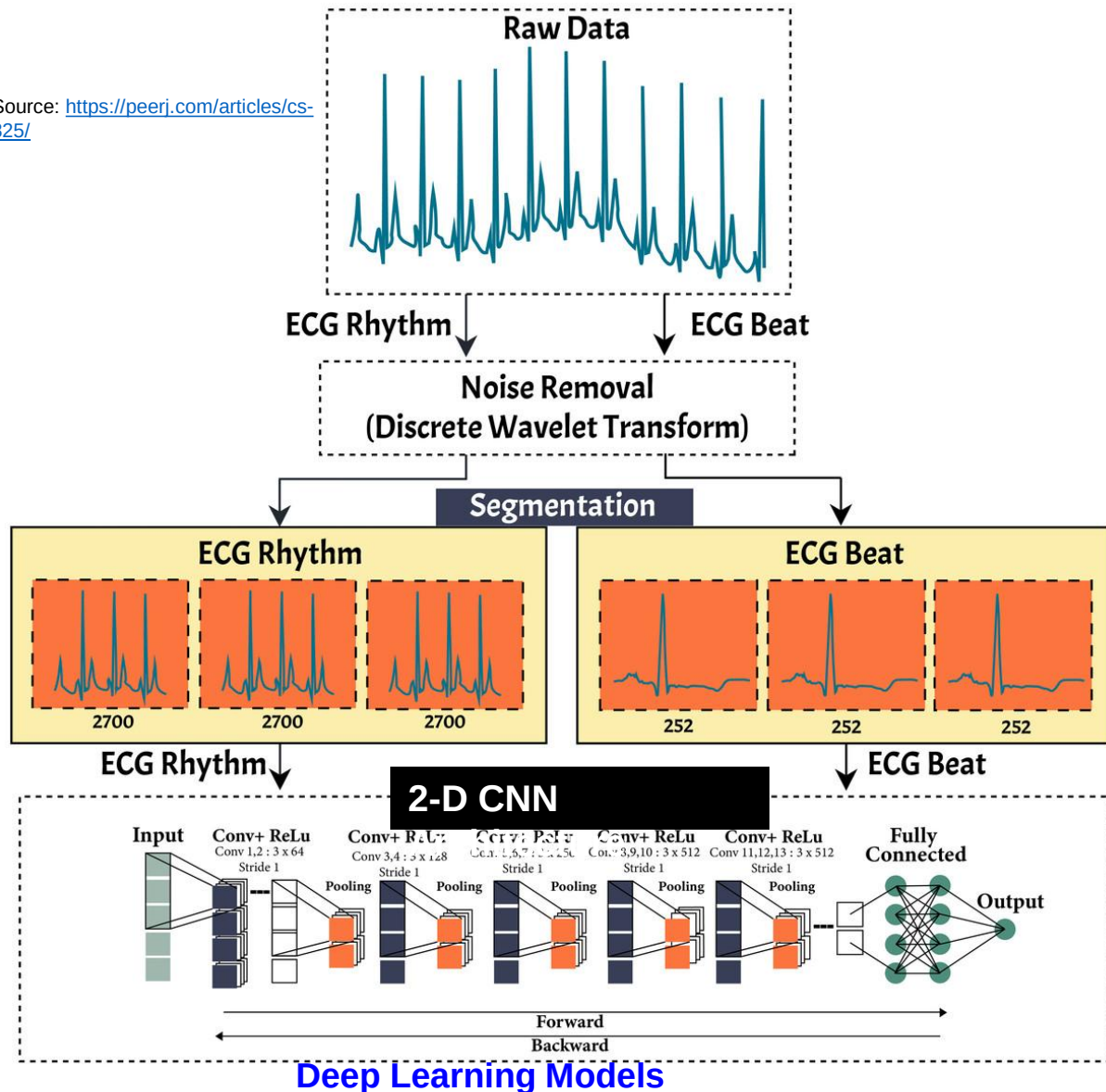
➤ **Traditional machine learning methods**, such as Support Vector Machine (SVM), Principal Component Analysis (PCA), K-Nearest Neighbors (KNN), and Hidden Markov Model (HMM), have significantly advanced ECG classification.

➤ These methods rely on pre-processing, manual feature extraction, and classification, achieving around **96% accuracy**.

➤ But in real-world use, **traditional ML algorithms** often **fail** with noisy or unbalanced data and are now seen as **outdated**.

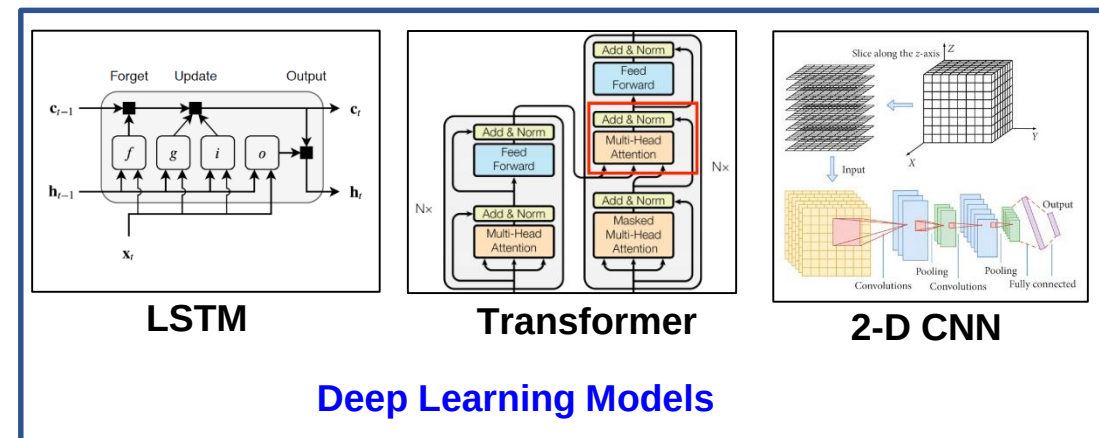
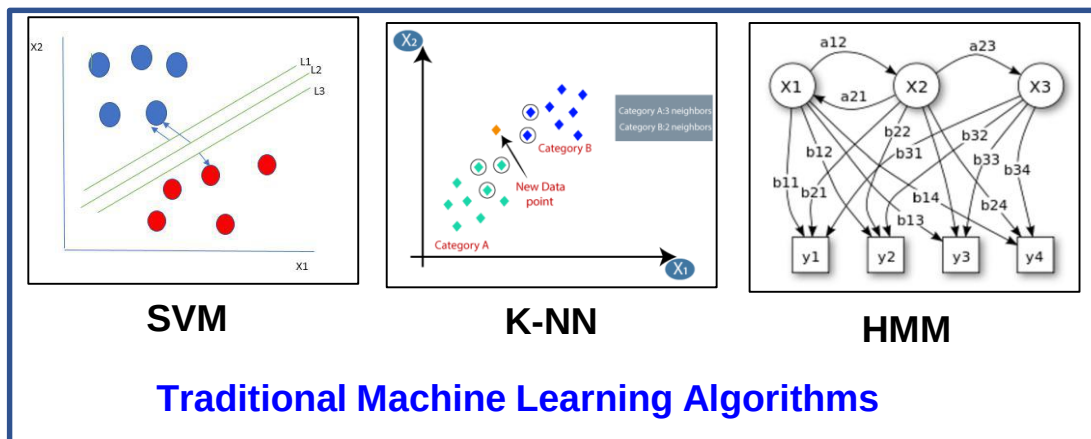
Background

Source: <https://peerj.com/articles/cs-825/>



- **Deep learning models** like Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and Transformers have outperformed traditional methods in ECG arrhythmia detection, achieving up to 99% accuracy.
- **LSTM** and **Transformer** designed to learn time-based patterns, work well with ECG signals but require too **much power and memory** for wearable devices.
- CNNs are more hardware-friendly, but **2D CNNs** still consume **high resources**, leading to research on lightweight FPGA-based solutions.

Background



Traditional ML and deep learning models are often **not suitable for wearable devices** due to high resource demands or limited accuracy.

Low
Accuracy

y

High Accuracy
but High
Resource

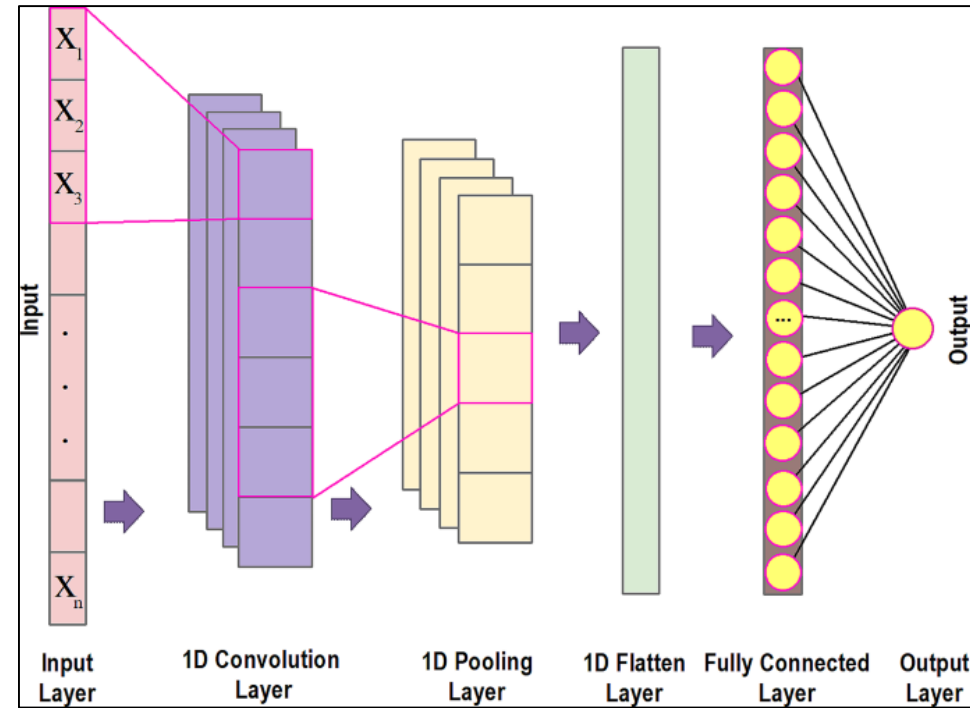


WEARABLE ECG



Background

Why don't we
apply **1-D CNN**
instead of 2-D
CNN for **ECG**
Classification?



1-D CNN

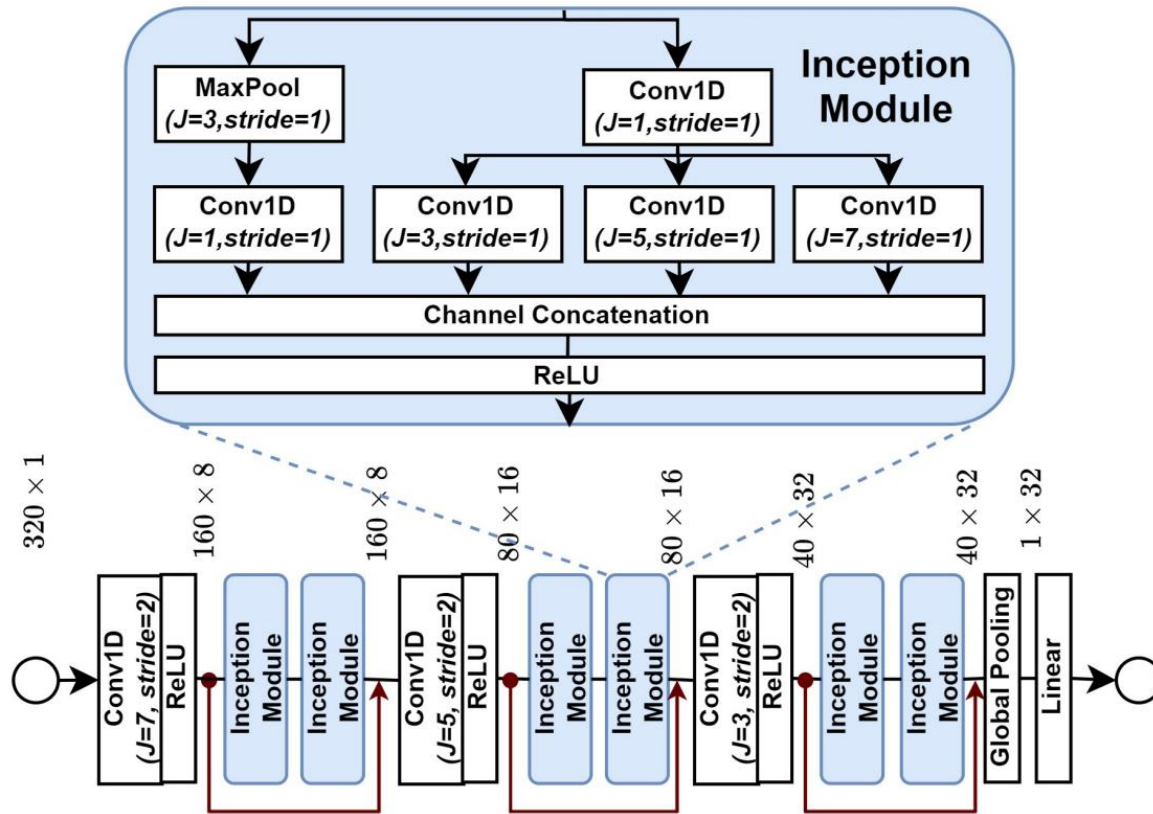
- ✓ **High Accuracy**
- ✓ **Low Resource**



WEARABLE ECG



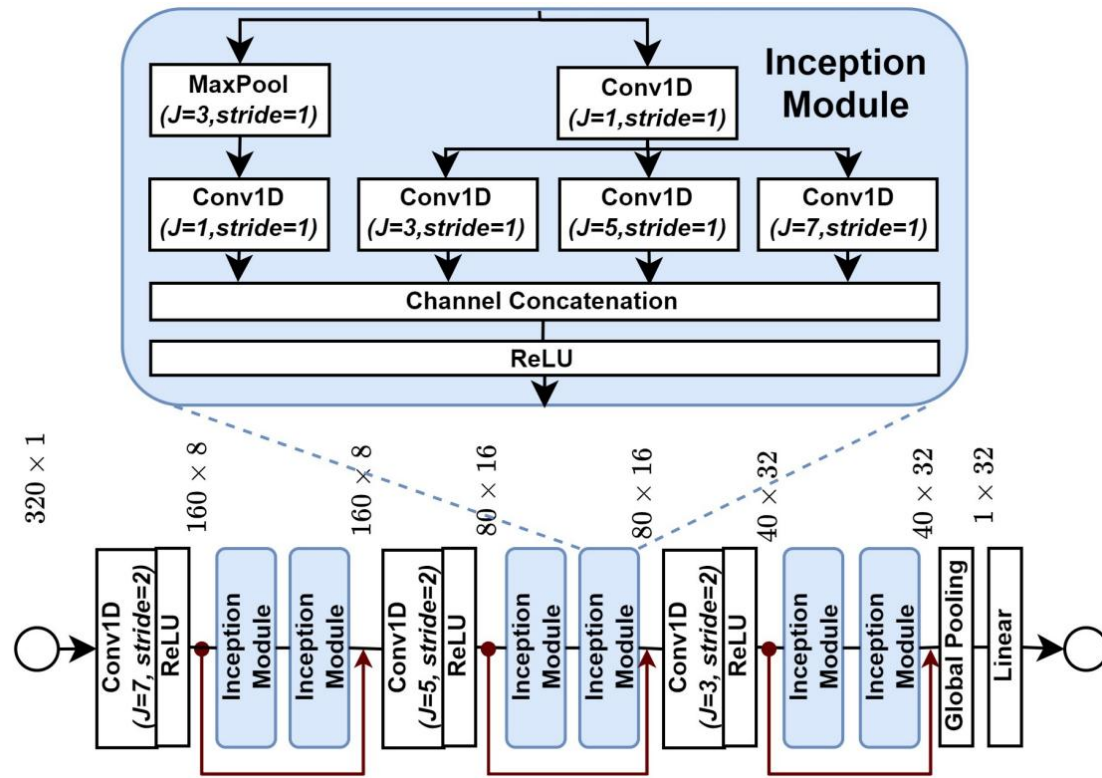
Proposed Accelerator for 1D CNN



The proposed Tiny InceptionNet model for ECG classification

- This work presents Tiny InceptionNet, a **lightweight** and **high-speed 1-D CNN model** for ECG classification.
- It uses an Inception Module with **parallel convolutions of different kernel sizes** (1×1 , 3×1 , 5×1 , 7×1) to capture multi-scale features.
- Inspired by InceptionNet, ResNet, and DenseNet, it applies software-level optimizations to **reduce the number of parameters**.

Proposed Accelerator for 1D CNN



The proposed Tiny InceptionNet model for ECG classification

✓ Reduce Parameter

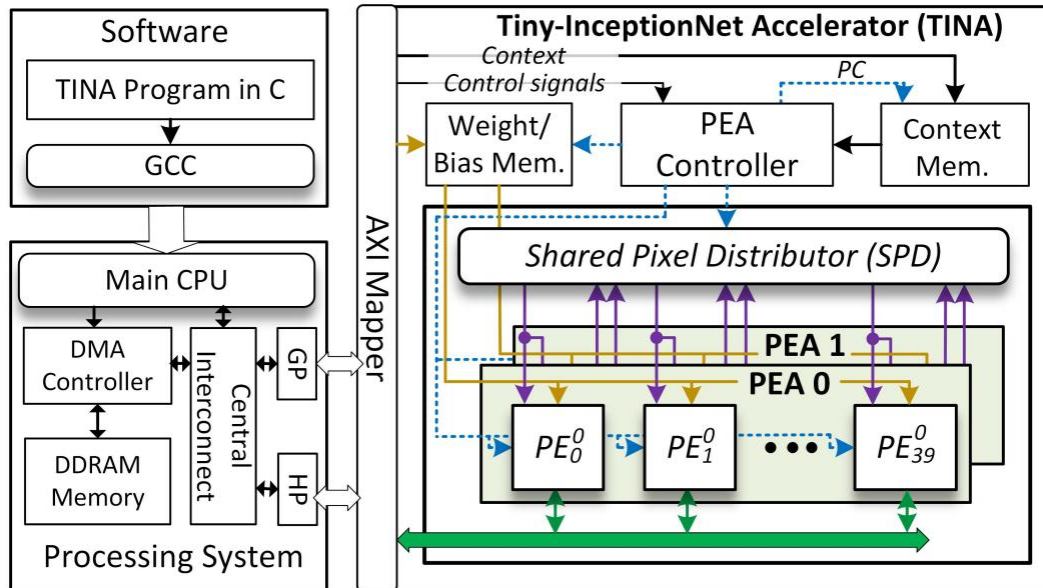
✓ Reduce Memory Usage

✓ High Accuracy

Accuracy & Area Optimization



Proposed Accelerator for 1D CNN



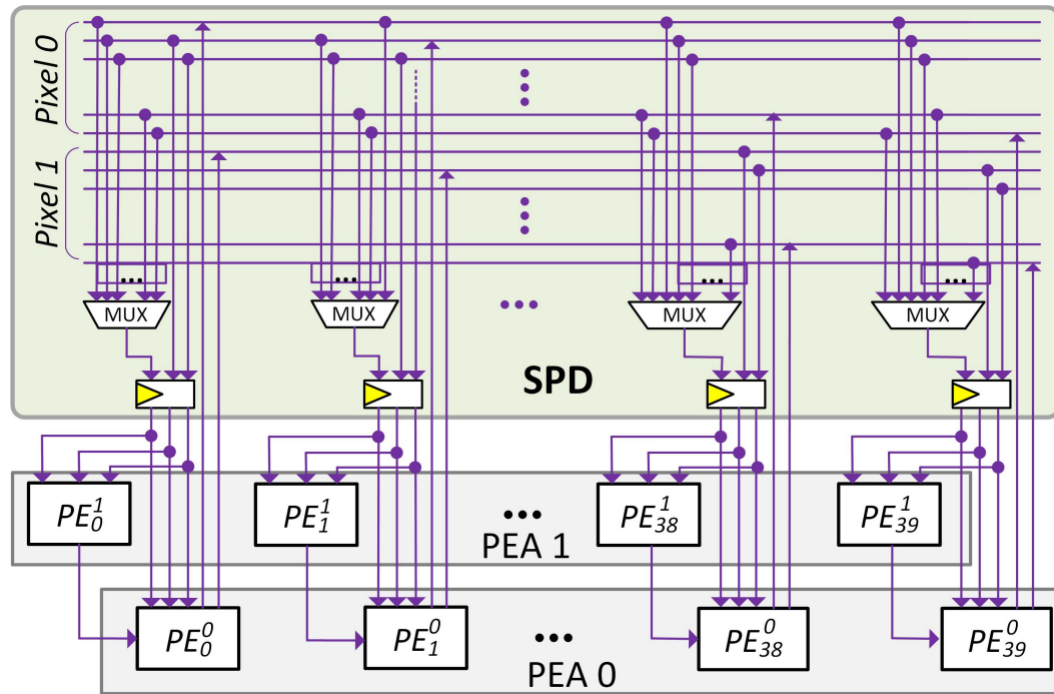
❑ The TINA SoC architecture includes the processing system (PS), TINA software, and TINA hardware.

❑ The TINA hardware features a **dual PEA**, consisting of PEA 0 and PEA 1, each containing 40 processing elements (PEs).

TINA overview architecture featuring a dual PEA at the SoC level

❑ PEA 0 handles conv, pooling, add, activation; PEA 1 handles conv only

Proposed Accelerator for 1D CNN



Simplified architecture of the shared pixel distributor (SPD)

- ❑ PEA 0 and PEA 1 perform parallel computation with **network topology parameters**, such as varying kernel sizes, strides, and channel dimensions.

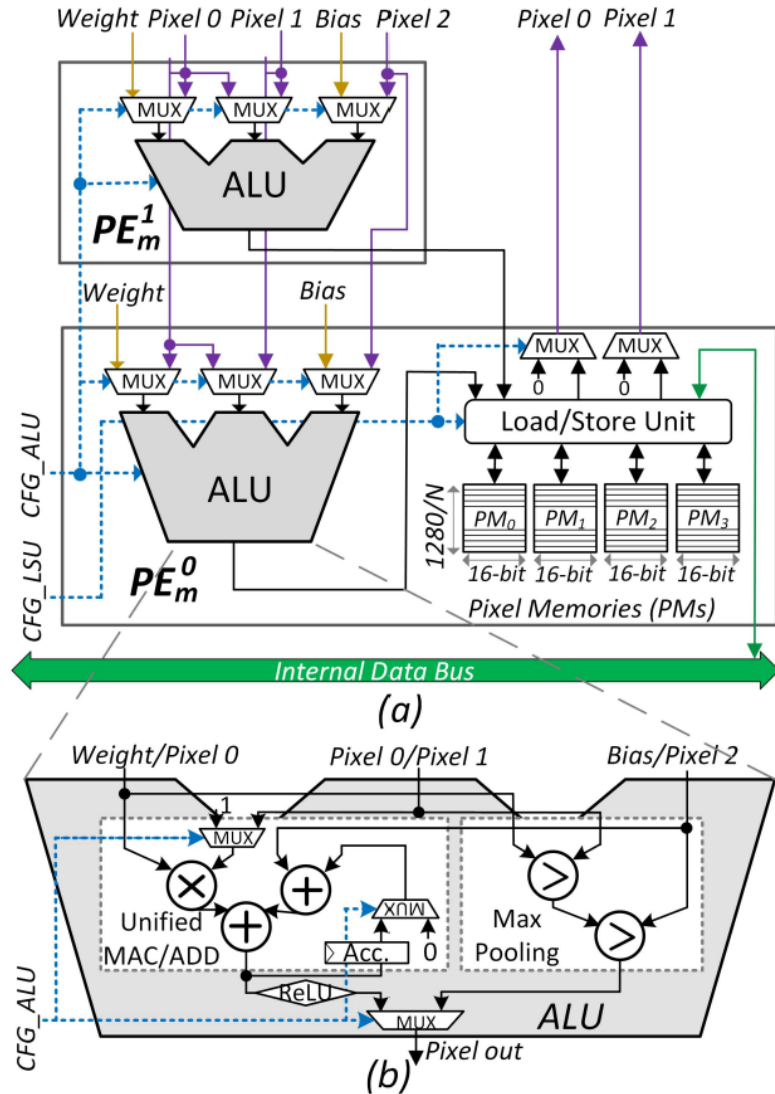


- ❑ Efficient pixel coordination is needed to optimize the pipeline and computation speed.



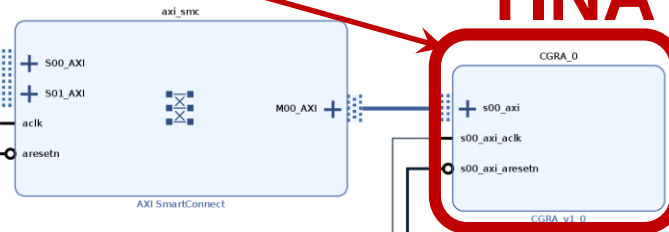
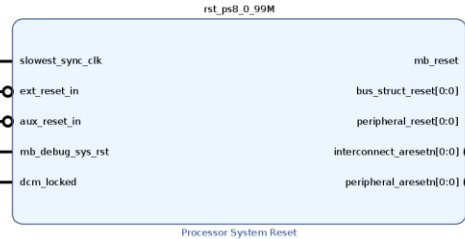
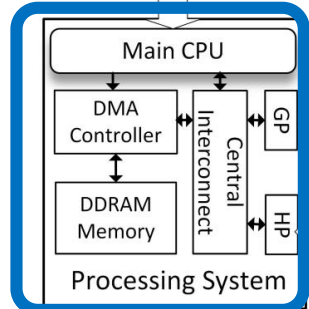
- ❑ We propose the **Shared Pixel Distributor (SPD)** to dynamically route input data to 80 PEs in the dual PEA.

Proposed Accelerator for 1D CNN



- ❑ To enable conflict-free memory access, PEs from dual PEA share a dual-port memory.
- ❑ The PE in Figure (a) includes an ALU, LSU, and four Pixel Memories (PMs) for efficient computation with minimal external memory access.
- ❑ Figure (b) illustrates a configurable ALU that supports MAC, addition, ReLU, and Max Pooling, with inputs selected via MUX and controlled by the CFG ALU signal for flexible neural network computations.

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TINA

Vivado tool



Result

COMPARISON OF PARAMETER COUNT AND ACCURACY FOR ECG CLASSIFICATION ON THE MIT-BIH DATASET WITH OTHER METHODS

Ref.	[7]	[8]	[9]	[10]	[12]	[15]	This work
Classifier	SVM	CNN-LSTM	2-D	2-D CNN	1-D CNN	1-D CNN	1-D CNN
#Parameter	-	-	1.16M	3.68M	11,065	-	6,261
ACC (%)	98.39	98.42	99.05	99.11	99.13	99.3	99.49
SEN (%)	96.86	98.07	97.85	97.91	99.13	99.3	99.49
SPEC (%)	98.92	98.76	99.57	99.61	98.59	-	98.87
PPV (%)	96.85	98.76	98.55	98.58	99.13	-	99.50

- ✓ Tiny InceptionNet achieves the **highest accuracy** and sensitivity (**99.49%**) among compared models.
- ✓ It uses only 6,261 parameters, **43% fewer** than the smallest existing model.



Result

Reference		[11]	[12]	[13]	[14]	[15]	[16]	This Work
Model		1-D CNN	1-D CNN	1-D CNN	1-D CNN	1-D CNN	1-D CNN	1-D CNN
Dataset		MIT-BIH	MIT-BIH	MIT-BIH	Chapman	MIT-BIH	MIT-BIH	MIT-BIH
FPGA		Artix-7	ZC706	ZC706	Cyclone V	PYNQ-Z2	ZC702	ZCU102 / ZC706
Clock (MHz)		25.5	200	200	50	10	150	250 / 200
CNN Size (GOP)		0.38×10^{-3}	1.03×10^{-3}	1.03×10^{-3}	0.38×10^{-3}	4.11×10^{-4}	—	0.32×10^{-3}
Parameter		9,385	11,065	11,065	-	-	-	6,261
Precision		22-bit Fixed	16-bit Fixed	16-bit Fixed	8-bit Fixed	8-bit Fixed	16-bit Fixed	16-bit Fixed
Area	LUT	128,960	1,538	2,510	33,870 [‡]	10,870	19,700	28,758 / 30,388
	BRAM	16.5	12	12	1	53	44	4.5 / 4.5
	DSP	121	96	96	44	53	73	80 / 80
	eLUT [†]	175,776	33,346	38,798	46,974	27,278	74,636	54,686 / 56,316
IT (μs)		1,358	64.25	53.54	66	233	676.20	17.80 / 22.25
ADP (s $\times$ eLUT)		238.70	2.14	2.08	3.10	6.36	6.36	0.51 / 0.68

[†]The eLUT is normalized by $\#LUT + \#BRAM \times 784 + \#DSP \times 280$.

[‡]The Adaptive Logic Module (ALM) in Altera Cyclone FPGAs is normalized to the LUT in Xilinx ZC706 FPGAs as $1 \text{ ALM} \approx 1.5 \text{ LUTs}$.

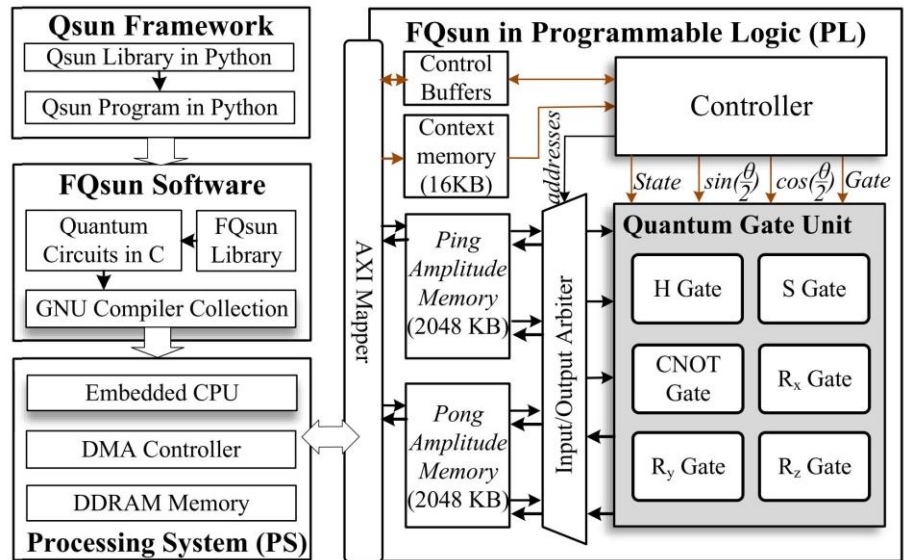
- ✓ TINA outperforms state-of-the-art FPGA-based 1-D CNNs with **2.4× faster inference time** and **3.06× better ADP**.



III. CGRA-based Accelerator for Quantum Emulation

Publication

Quantum Emulator on FPGA



[HPQEA: A Scalable and High-Performance Quantum Emulator with High-Bandwidth Memory for Diverse Algorithms Support](#)

T Van Duy, TH Vu, VTD Le, HL Pham, Y Nakashima
arXiv preprint arXiv:2510.07110

[Transfer-Based Strategies for Multi-Target Quantum Optimization](#)

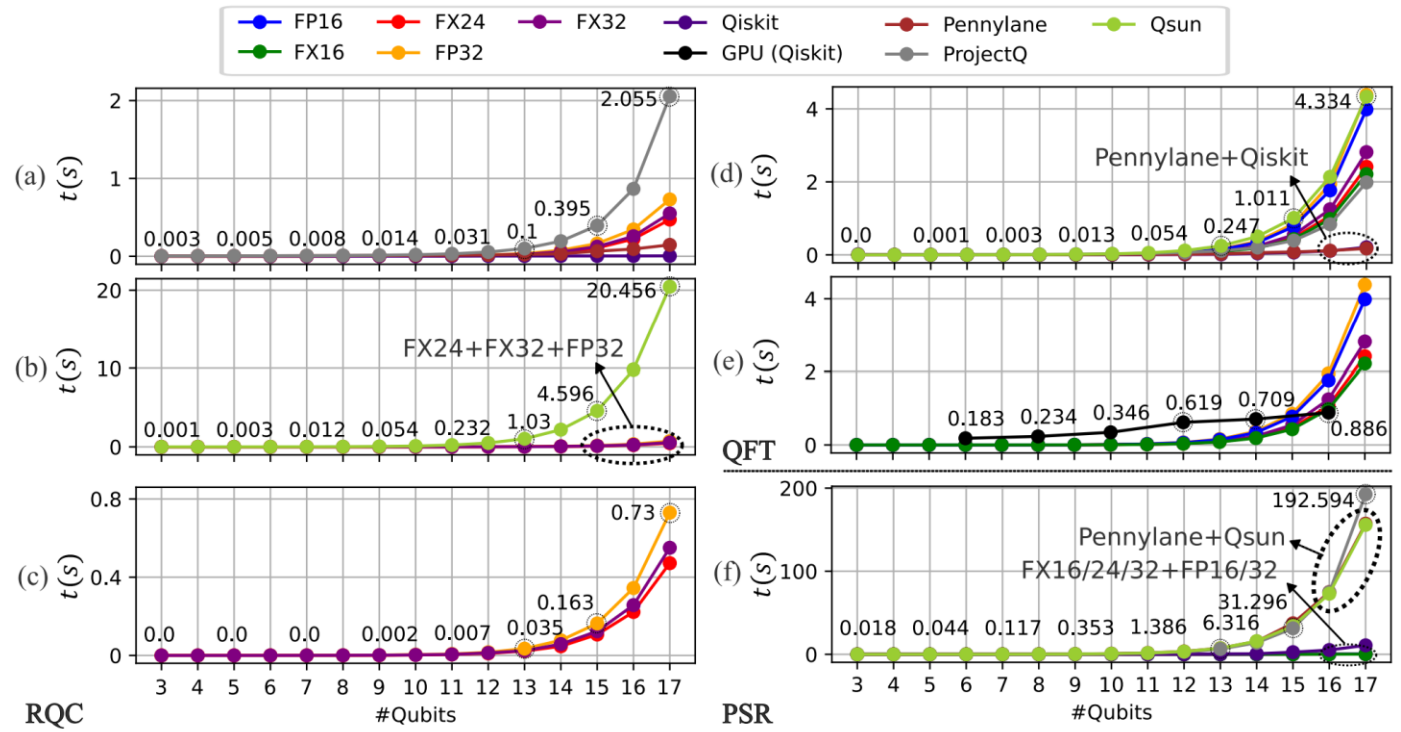
VT Hai, BC Doanh, LVT Duong, PH Luan, Y Nakashima
arXiv preprint arXiv:2508.11914

[Benchmarking variants of the Adam optimizer for Quantum Machine Learning Applications](#)

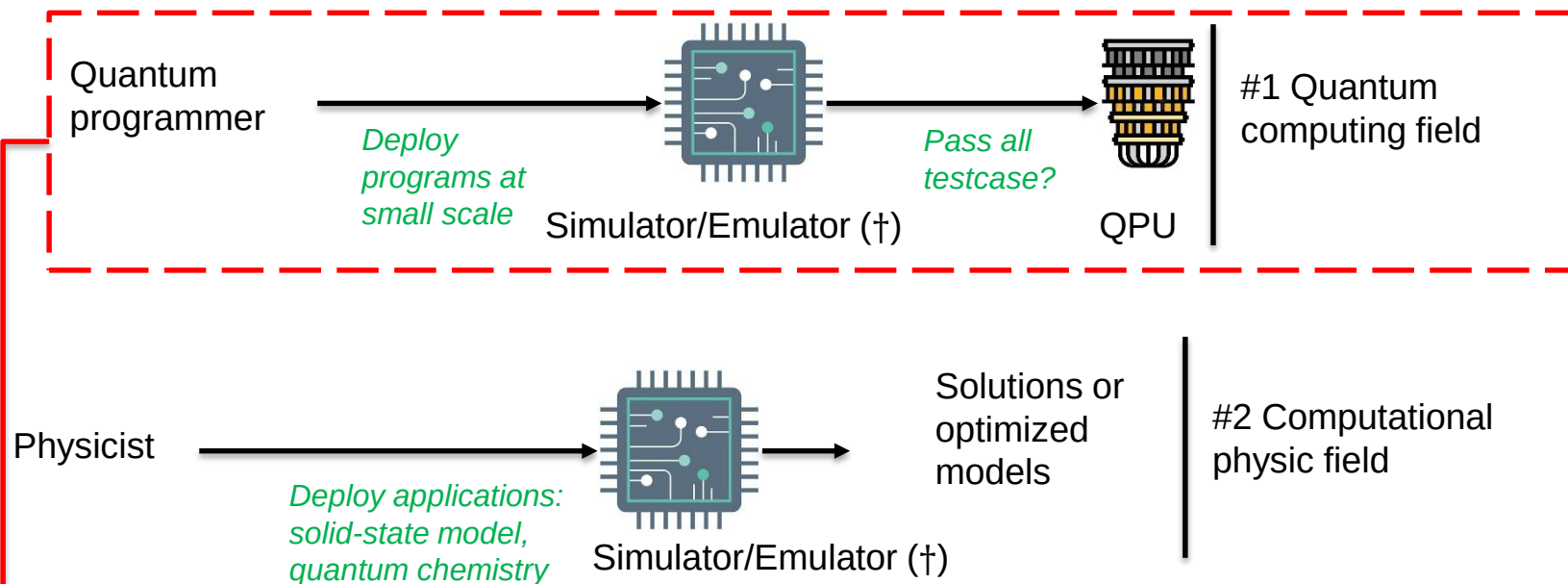
TH Vu, VTD Le, HL Pham, Y Nakashima
IEEE Open Journal of the Computer Society

[QEA: An Accelerator for Quantum Circuit Simulation with Resources Efficiency and Flexibility](#)

TH Vu, VTD Le, HL Pham, Y Nakashima
2025 10th IEEE International Conference on Integrated Circuits, Design, and ...



Background

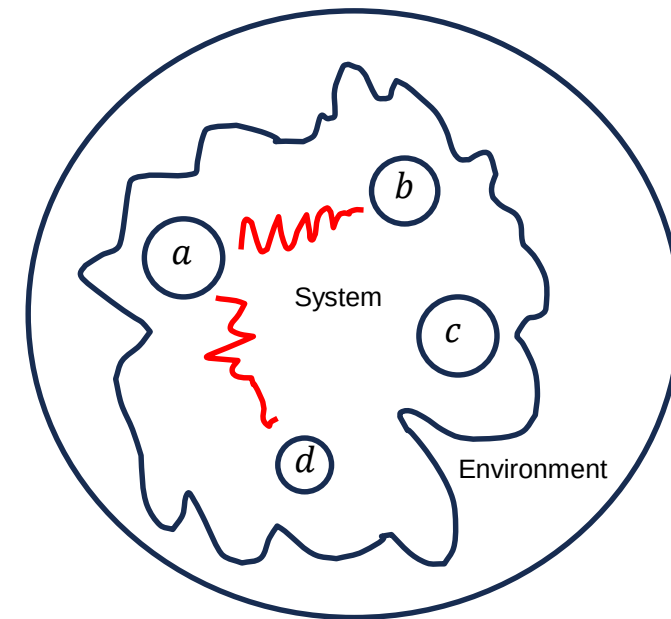


We focus on the **usecase #1**, but the designs are applicable to the **usecase #2**

Target: Solve the Equation (0.1) to meets the demands of speed, flexibility, and energy efficiency

(†) Quantum simulator: run on CPU/GPU

Quantum emulator: run on other devices (FPGA, ...)



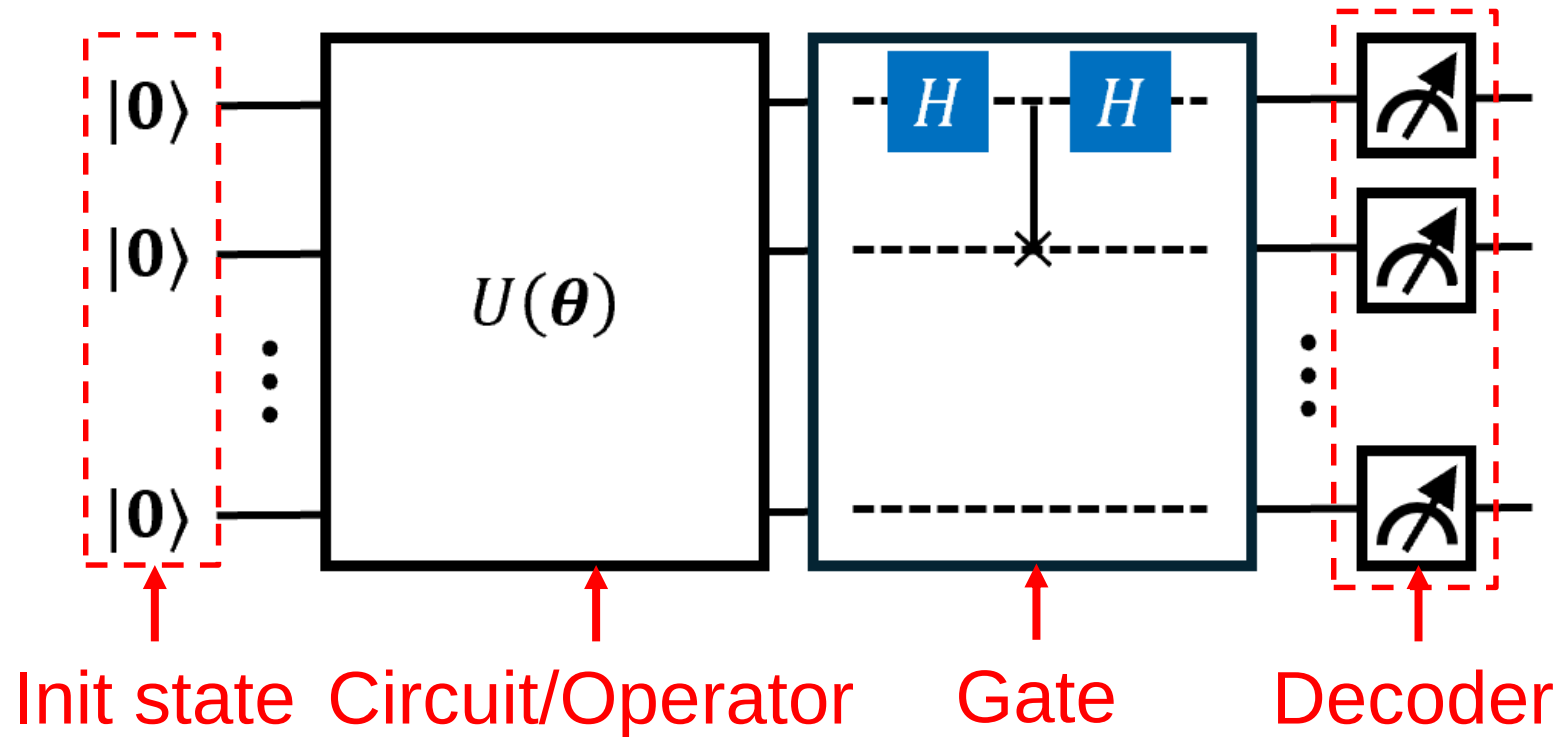
Multiple-particle (closed) quantum system.

Schrodinger's equation for describing system [1]:

$$i\hbar \frac{\partial \Psi(\vec{r}_1, \vec{r}_2, \dots, t)}{\partial t} = \hat{H} \Psi(\vec{r}_1, \vec{r}_2, \dots, t) \quad (0.1)$$

[1] Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information*. Cambridge university press.

Background



A n – qubit quantum circuit.

- ❑ A quantum gate (gate) $g(w, \theta)$ act on single wire (1-qubit) or multiple wires (2^+ -qubit) w with parameter θ (if applicable).
- ❑ Decoder (Observable) return the measurement values.

Background

Surveyed results of related quantum simulators/emulators in the last five years.

(†) Depend on hardware resources

(††) Estimated **ONLY**

(†††) Best performance in 3-17 qubits.

Reference	Devices	Precision	#Qubits	Metrics
Qiskit	CPU/GPU/QPU	FP32,FP64	(†)	Exe time
PennyLane	CPU/GPU	FP32,FP64	(†)	Exe time
Analog-based emulator	UMC-180 nm CMOS analog	FP32	6-17	Exe time, Power consumption
ZCVU9P-based emulator	Xilinx ZCVU9P	FX16	6	Exe time, test accuracy
Alveo-based emulator	Alveo U250	FP32	32 (††)	Exe time, resources utilization
Arria-based emulator	Arria 10AX115N4F45E3SG	FX16	32 (††)	Exe time, resources utilization, frequency
XCKU-based emulator	Xilinx XCKU114	FX16	16	Exe time, resources utilization
Stratix-based emulator	Altera Stratix EP1S80B956C6	FX8, FX16	8	Exe time, logic cell usage, frequency
This work (FQsun)	Xilinx ZCU102	FX16/24/32, FP16/32	30 (†††)	Exe time, PDP, Fidelity, MSE, Power, Resources utilization

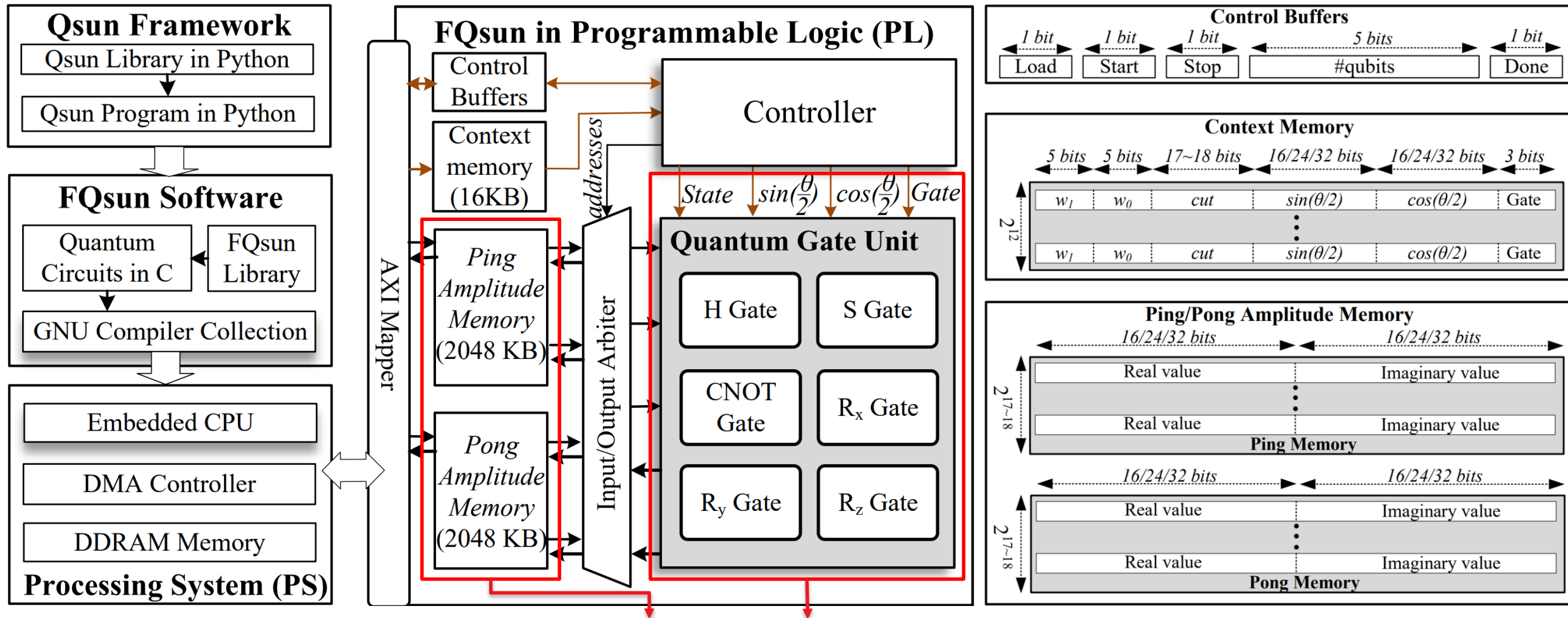


Universal simulator/emulator



Limited (Fixed) emulator

Proposed Quantum-Emulating Accelerator

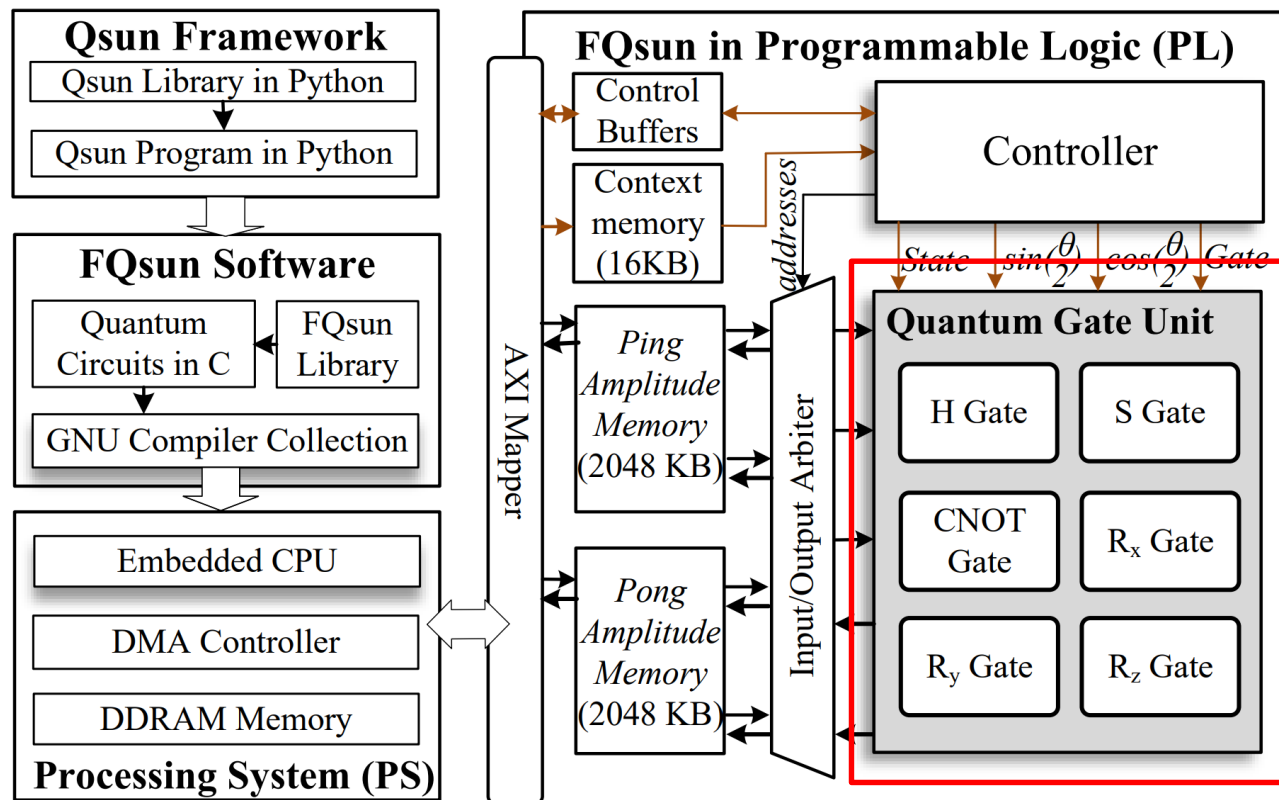


Overview architecture of our FQsun at the system-on-chip (Soc) level on FPGA.

Ping-Pong memory **QGU**

Memory organization of the proposed FQsun.

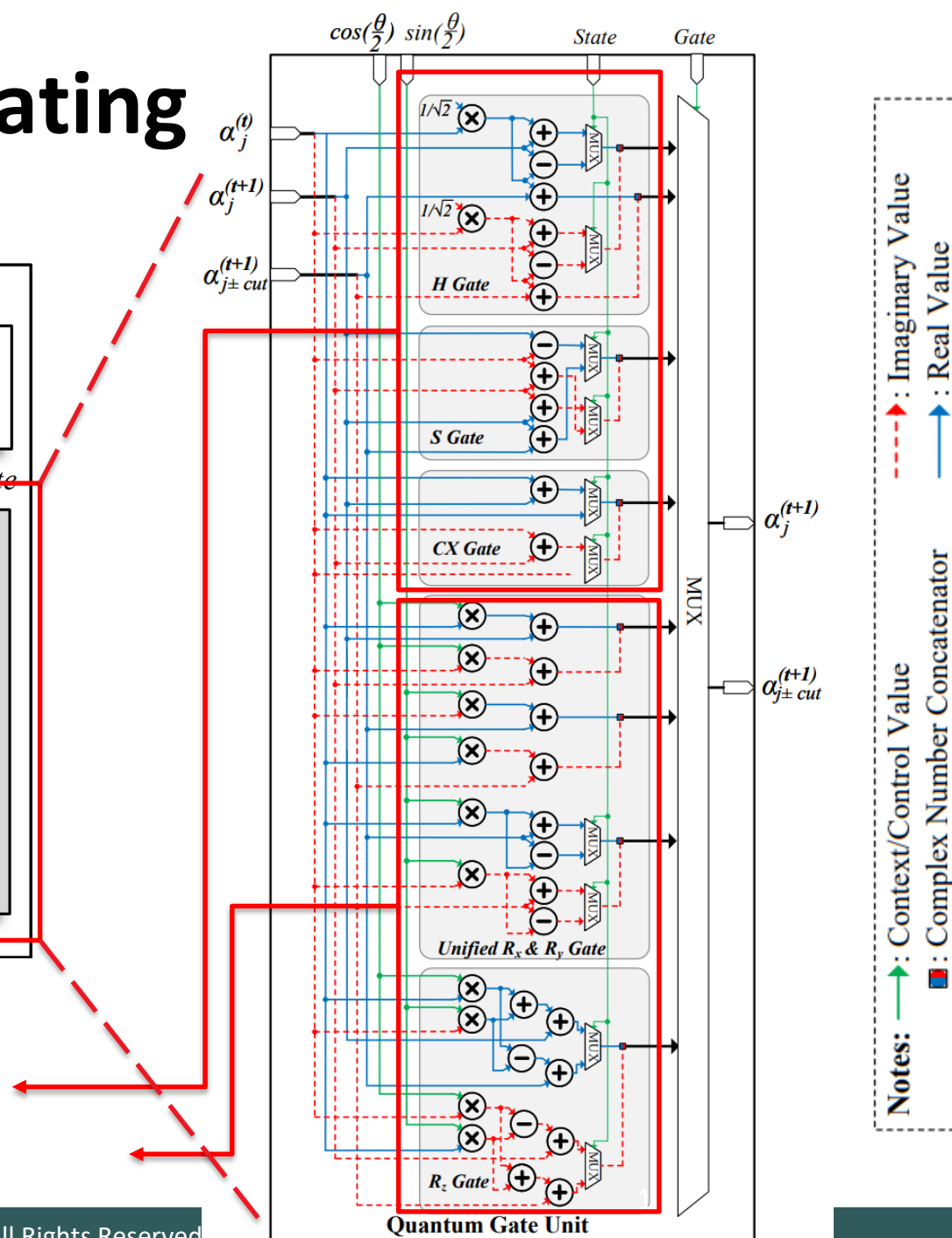
Proposed Quantum-Emulating



Micro-architecture of Quantum Gate Unit (QGU).

Clifford gates: H, S, CX

Non-Clifford gates: R_x, R_y, R_z



Result

Comparative analysis of FQsun and existing FPGA-based emulators on QFT's performance.

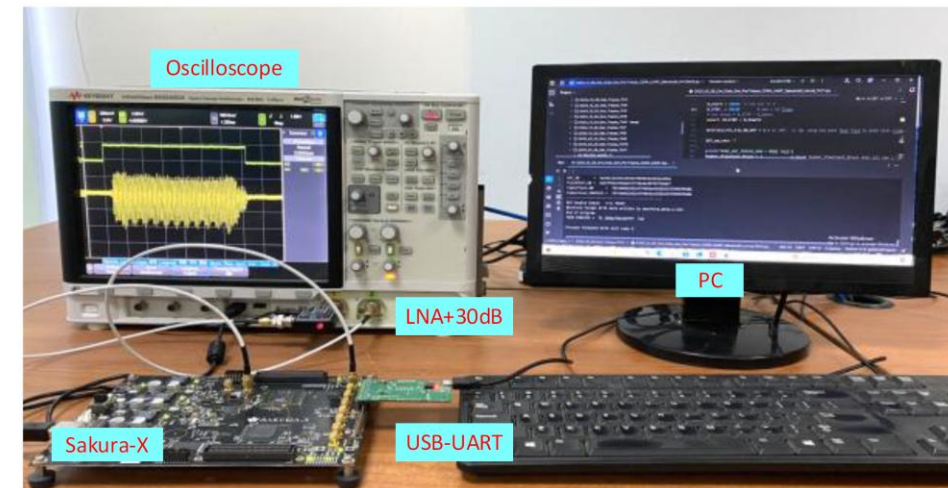
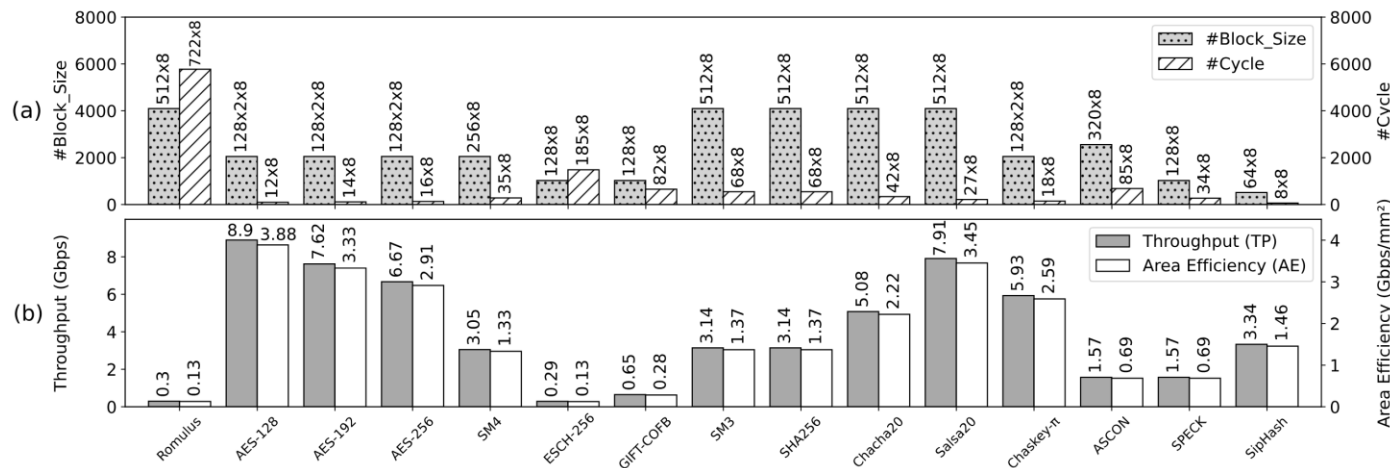
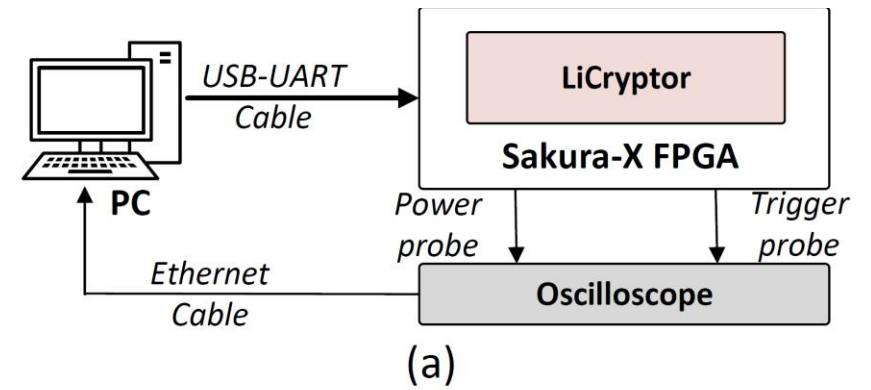
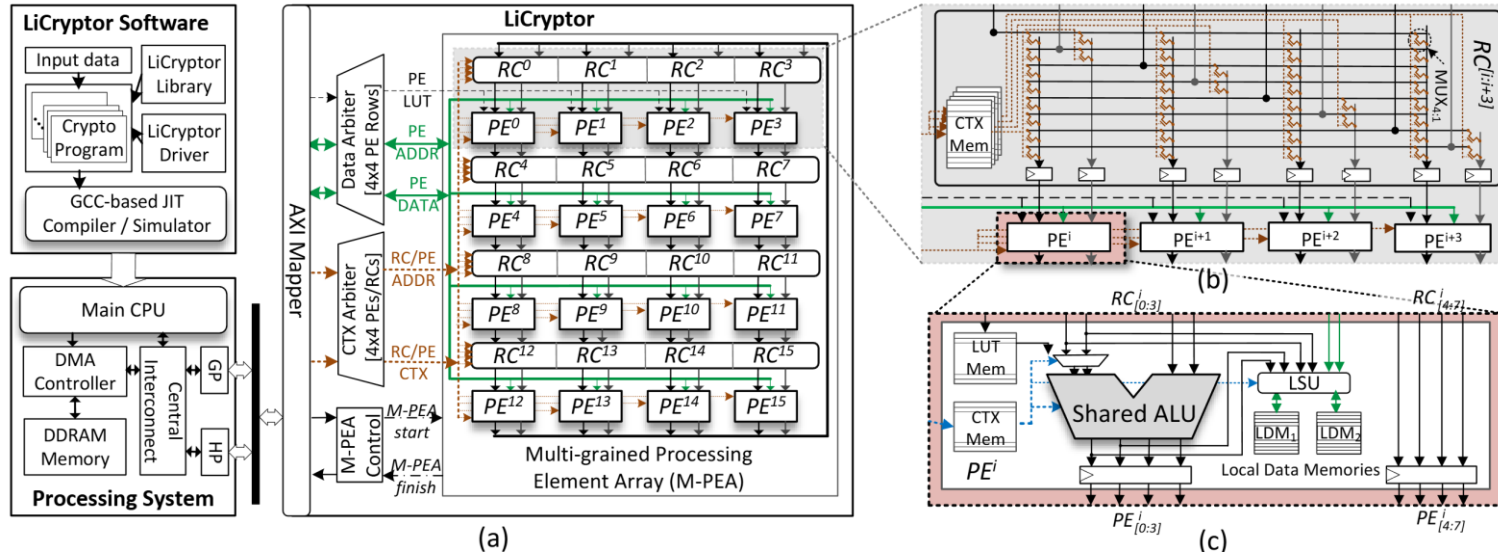
Works	Device	Frequency (MHz)	Flexibility	Precision	#Qubits	Execution time (s)	#Gates [†]	NGS ^{††}
[57]	Xilinx XCVU9P	233	✗	18-bit FX	16	1.20×10^{-3}	-	-
[2]	Arria 10AX115N4F45E3SG	233	✓	32-bit FP	16	1.84×10^1	528	5.33×10^{-7} No flexibility
[59]	Xilinx XCKU115	160	✗	16-bit FX	16	2.70×10^{-1}	136	3.03×10^{-8}
[63]	AMD Xilinx Zynq-7000	100	✓	32-bit FX	6	1.15×10^{-4}	10	1.80×10^{-7}
[64]	2 × Intel Stratix 10 MX2100	299	✗	32-bit FP	30	4.47×10^0	465	8.95×10^{-12} No flexibility
This work	Xilinx ZCU102	150	✓	16-bit FP	18	8.90×10^0	810	4.19×10^{-8}
		136	✓	32-bit FP	17	4.30×10^0	721	4.55×10^{-8}
		136	✓	16-bit FX	18	4.90×10^0	810	2.31×10^{-8}
		125	✓	24-bit FX	17	2.41×10^0	721	2.52×10^{-8}
		125	✓	32-bit FX	17	2.81×10^0	721	2.97×10^{-8}



Current Research Direction

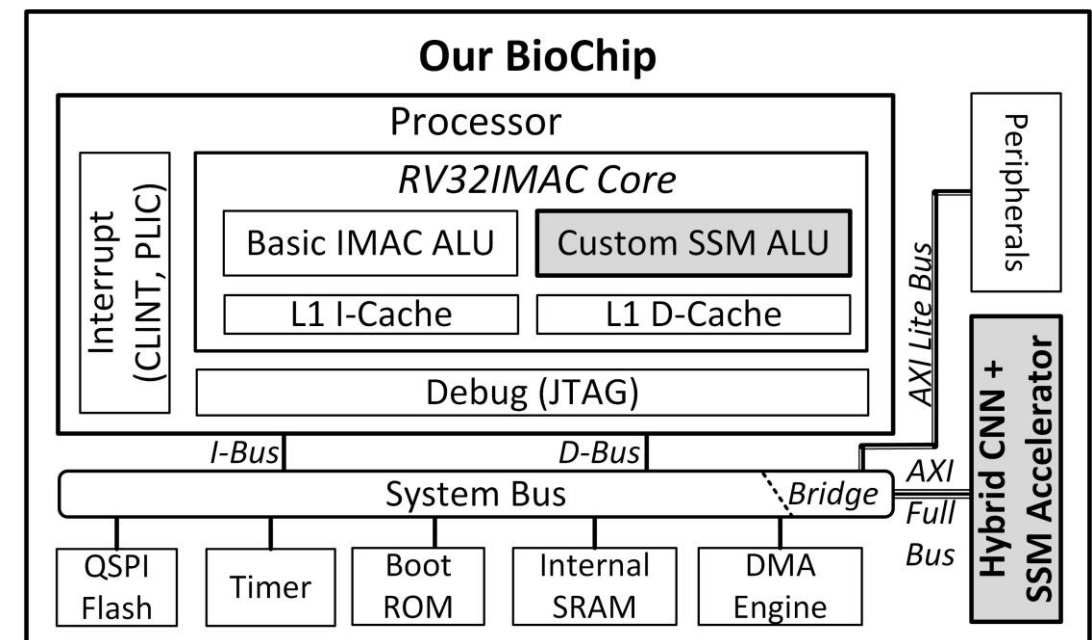
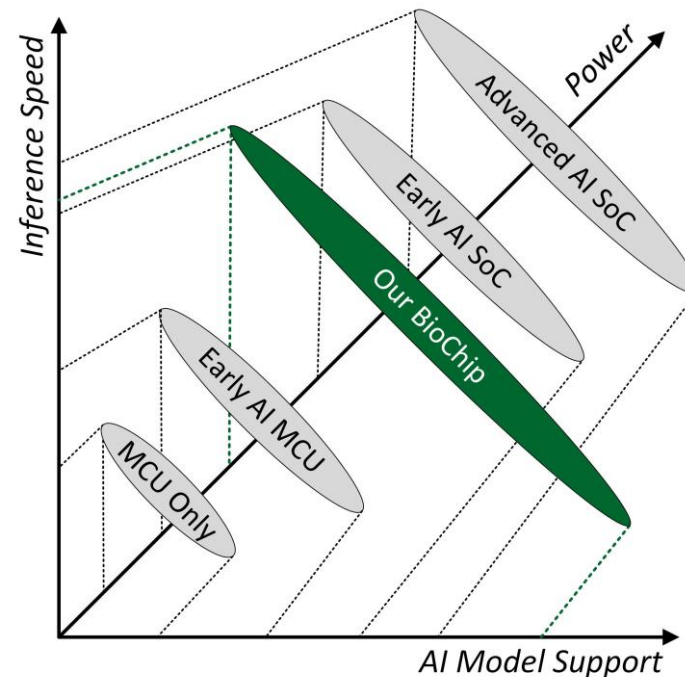
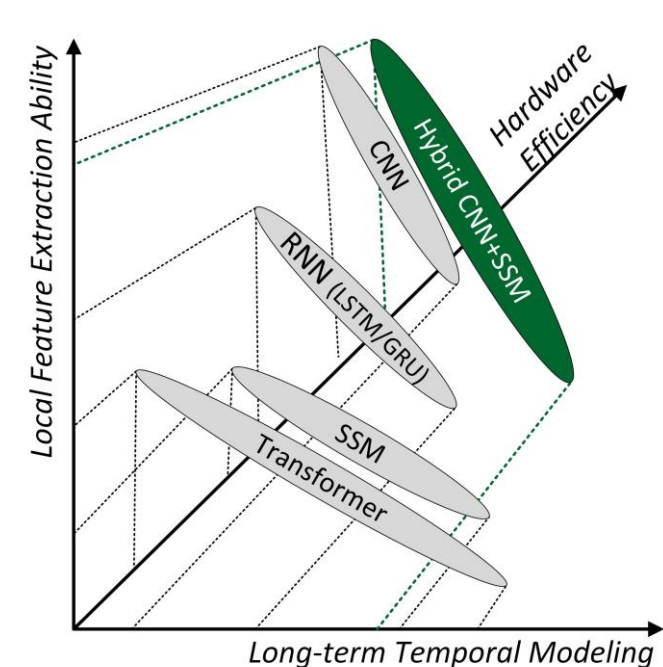
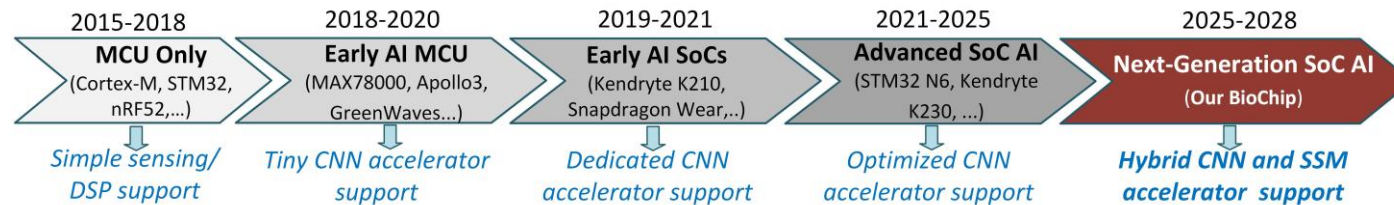
Current Research Directions

Side-Channel Attacks and Defenses for CGRA-Based Cryptographic Accelerator



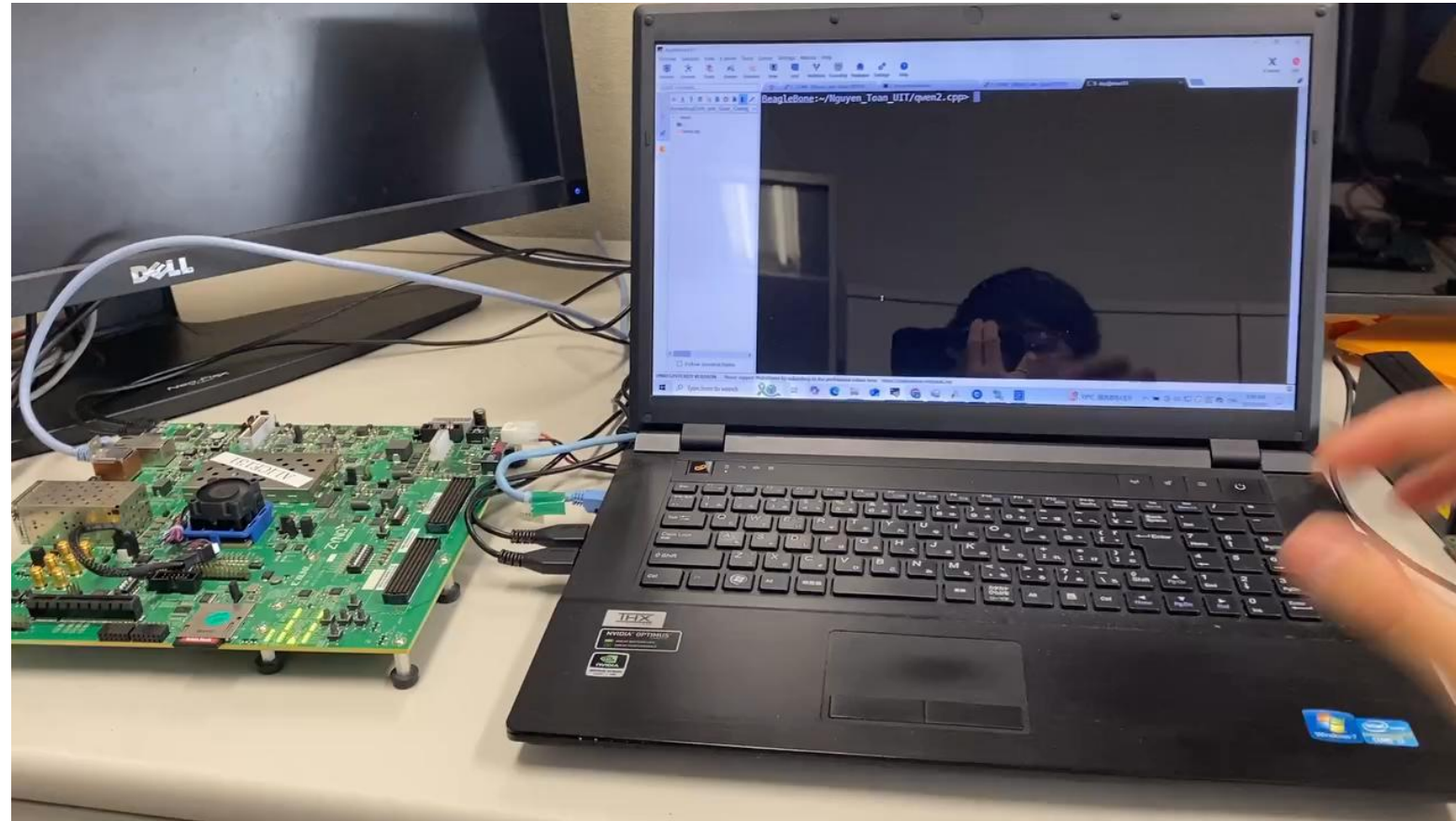
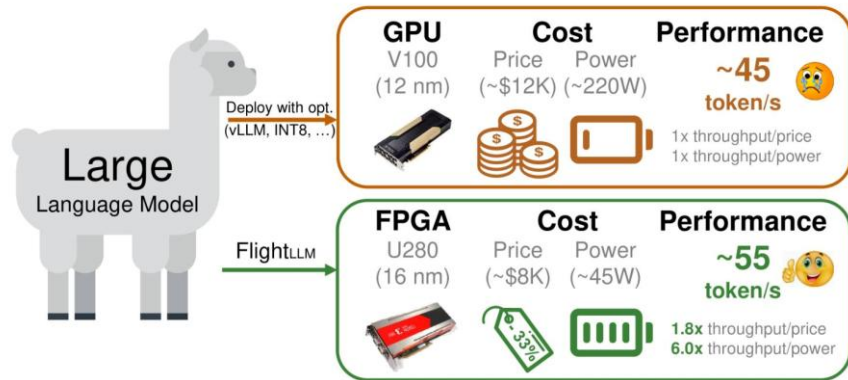
Current Research Directions

SoC with Deep Neural Accelerator for Biosignal Processing

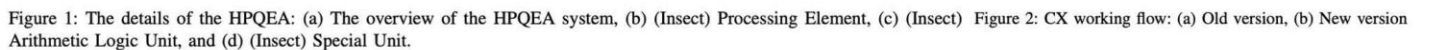


Current Research Directions

❑ FPGA for SLM/LLM Acceleration



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Q&A



Thank
you